AI-Based Indoor Localization and Tracking for IoT Networks

By

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Project Title: AI-Based Indoor Localization and Tracking for IoT Networks

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Abstract

The progress of wireless technologies together with Internet creation has made it possible to establish IoT as a fundamental part of human life. Among all service applications in these systems the localized services stand out as the most intriguing implementation. Target positions become detectable through deployments of sensor networks which gather and process sensor data. Researches have established different localization systems which are described in published literature. Multiple localization systems implement distinctive positioning methods supported by different unique technologies and approach techniques. The range of system capabilities allows users to select those solutions which match their exact needs more effectively. This document presents an extensive breakdown of localization operations in Wireless Sensor Networks (WSNs) by exploring diverse main localization methods that operate using different technological implementations. Research shows that localization systems need to be categorized through their approach into interactive systems distributed systems and centralized systems. The techniques that make up localization systems belong to three categories because they include measurement of distance and angles and measurements of areas and hop-count-based tracing.

Contents

List of Abbreviations

AI – Artificial Intelligence

IoT – Internet of Things

WSN – Wireless Sensor Network

RSSI – Received Signal Strength Indicator

ToF – Time of Flight

TDoA – Time Difference of Arrival

AoA – Angle of Arrival

GPS – Global Positioning System

UWB – Ultra-Wideband

BLE – Bluetooth Low Energy

Wi-Fi – Wireless Fidelity

LoRa – Long Range

RTT – Round Trip Time

CNN – Convolutional Neural Network

SVM – Support Vector Machine

LPWAN – Low-Power Wide-Area Network

NB-IoT – Narrowband Internet of Things

SLAM – Simultaneous Localization and Mapping

E2EE – End-to-End Encryption

AES – Advanced Encryption Standard

RSA – Rivest-Shamir-Adleman (Encryption Algorithm)

IMU – Inertial Measurement Unit

ML – Machine Learning

MQTT – Message Queuing Telemetry Transport

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Chapter 1: Introduction

1.1 Introduction to the Internet of Things (IoT)

IoT stands for Internet of Things which leads to technological interaction between connected physical devices in an internet-based data communication system. The development of IoT started from its inception by uniting different technologies including wireless sensor networks (WSNs) cloud computing artificial intelligence and big data analytics. The fundamental principle behind IoT enables different objects including household devices industrial machines medical equipment and automotive vehicles to link and execute self-operational tasks through network communication.

The deployment of Internet of Things technology gained extraordinary speed during the past 20 years. Its growth speed has been enhanced through technological advances in miniaturization and wireless communication together with high-speed internet. The global number of IoT-connected devices will surpass 75 billion by 2025 as they continue to transform numerous industrial fields across multiple sectors. IoT technology lets healthcare practitioners view patient vital signs remotely so they can intervene promptly for superior medical care. The real-time tracking offered in logistic operations cuts down losses while allowing better control of inventory. Cities using IoT technology improve energy performance and reduce gridlock while delivering enhanced quality of living to residents. The widespread applications demonstrate how IoT produces better operational efficiency and safety mechanisms as well as decreased costs throughout all sectors.

Many IoT applications succeed because they need precise location identification of networked devices. The functionality and reliability of various IoT applications depend heavily on their ability to perform accurate localization because it enables the tracking of supply-chain objects and positions self-governing vehicles and smart building sensor detection. Different operational needs and environments have led to the creation of various positioning methods through increasing development efforts.

Position estimation remains essential for wireless sensor network applications operating in monitored areas including animal habitats and aquatic observations and agricultural domains as described in [2]. The provision of location data supports multiple new application systems that consist of intrusion detection and inventory management and traffic tracking and healthcare applications among others [2]. The positioning approaches adopt geometrical measurement principles using triangulation and trilateration approaches as defined in [3] and [4]. Several methods exist to measure two node distance through synchronization radio signal strength and physical characteristics of the carrying wave [5].

The WSNs allow localization techniques to operate effectively through inter-measurements in the network which include distance, time difference of arrival and connectivity and angle of arrival without prior position knowledge after obtaining just a few locations [5]. A localization information sensor becomes an anchor when it possesses knowable prior location information.

Research studies effectively handle all requirements which sensor network localizations need to fulfill. GPS systems or positioning protocols cannot be used together with even if-configuration in this scenario because of excessive energy usage alongside their GPS dependencies. GPS-based services have a specific drawback since they perform inadequately inside enclosed spaces. A suitable solution involves using numerous sensors spread across the area while lowering the number of locations with known sensor placement to include GPS equipment. The majority of anchor-based techniques explored in literature utilizes Bayesian Monte Carlo methods per [7] to generate solution coverage particles yet they suffer from significant memory limitations.

The evolution of technology produces enhanced localization capabilities despite the superior convenience of device free systems in diverse essential applications. The system enters recording mode when an abnormal object intrudes due to the shared functionality between intrusion detection and tracking processes [8]. A device-free application exists to monitor the safety of independent disabled individuals and senior citizens in stroke and fall emergency situations. Market expansion necessitates the creation of a position-giving device-free tool that meets all requirements at affordable costs. Device-free systems have unique characteristics that create new challenges for locating indoor positions. This paper examines critical factors affecting local localization methods as well as their technological implementations and associated techniques. The research examined present localization systems to determine solutions for destichotel and discernible detection through UWB radar and ultrasound devices alongside centralized and distributed iterative localization methods. Following all discussions in this paper a detailed performance analysis will be provided about multiple systems that make up location-based solutions. The taxonomical representation of wireless indoor localization.

This particular survey structure exists in the survey's second portion. providing an overview of all methods used for localization techniques within Wireless Sensor Networks. dealing with various techniques of localization. The analysis describes two types of localization technologies: both device-free and device-based systems. Section5 encompasses methods of localization and their IoT application. The study investigates pending research challenges as well as prospective study paths for this emerging field that researchers must examine.

In the present-day world of technology, IoT has come into being as a revolutionary transformer, enabling the seamless communication and interaction of devices across networks. One of the most crucial areas in this technological ecosystem is signal localization, that is, to determine the physical location of a device given its communication signals. Dropping the fields of mass scale rolling in the industries of healthcare, transportation agriculture, logistics, and smart cities will enable the importance of accurate and real-time localization to multiply. This paper delves into the need for, enabling technologies, and actual applications and challenges in signal localization in the IoT paradigm.

How IoT Signal Localization Works

IoT signal localization refers to all the specifiers and technologies that delineate the geographical or spatial position of IoT devices by means of the signals that these transmit or receive. Signal types include radio frequency (RF), Bluetooth, Wi-Fi, GPS, Ultra-Wideband (UWB), ZigBee, LoRa, and so forth. Localization is broadly classified into:

1- Outdoor Localization-Usually based on GPS signals or cellular signals.
2- Indoor Localization-As in WiFi fingerprinting, BLE, or UWB.

Localization methods can be divided again into range-based (Time of Arrival-TOA, Time Difference of Arrival-TDOA, Received Signal Strength-RSS, and Angle of Arrival-AoA are some metrics used) and range-free (e.g., centroid algorithms and fingerprinting).

Why Signal Localization Would Matter in IoT

1- Asset and Inventory Tracking

IoT signal localization enables real-time tracking of inventory and assets in storage and logistics industries. This ensures reduction of losses, a more efficient supply chain, and overall transparency of operations. In this respect, sensors fixed onto a package or equipment can relay its location so that managers can thereby automate inventory management and immediately respond to any issues.

2- Smart Cities

Localization is an essential factor in different applications of smart cities. Consider a traffic monitoring system or smart lighting all the way to waste management or even public transport; all have accurate location for the devices or for infrastructure components. Buses with IoT GPS-enabled devices, for example, would be able to provide real-time updates to the passengers, whereas smart bins would alert the collection agency when they needed emptying.

3- Emergency Safety

They provide human localization inside hazardous environments such as mines, oil rigs, or nuclear facilities under IoT signal localization. In case of an accident, emergency responders could quickly locate and provide assistance to the affected individuals. Furthermore, after a natural disaster or fire in a smart building, localization from wearables or smartphones can help rescue teams ascertain the location of the affected persons to evacuate and rescue them faster.

4- Precision Agriculture

Modern agriculture employs IoT for methods of monitoring soil moisture, crop health, and livestock behaviour. Localization of sensors and equipment should enable the farmers to apply them exactly where needed. GPS-guided autonomous tractors and drones rely on real-time localization data to maneuver effectively over huge farmlands.

5- Healthcare and Elderly Monitoring

In the healthcare sector-in remote patient monitoring or elderly care-wearable IoT devices track patient movement and location. Signal localization assures

1.2 Research Problem Threshold

1- The main issue when localizing IoT devices comprises simultaneous requirements for precise positioning and energy efficiency and affordable pricing. The indoor environments require methods other than GPS since GPS signals face severe signal degradation. Real-time operation is not achievable with machine learning-based localization approaches because these techniques need both substantial datasets and powerful computers.

2- The basic issue in IoT localization is the achievement of accuracy in real-time positioning while still ensuring minimum energy consumption and cost-effectiveness.

3- Conventional GPS-based solutions do not work in indoor environments due to signal attenuation and multipath effects. Furthermore, machine-learning-based localization approaches require very large datasets and high computational power, which hinder real-time implementation. Thus, it is essential to find an integrated multi-technology-based approach that compromises between accuracy, computer efficiency, and affordability: here lies the main research-gap.

1.3 Research Objectives:

The main objectives of this project entail:

- Testing the Performance of RSSI Combined with ToF and TDoA and AoA and Machine Learning Algorithms Dates back to the project to determine which combination works best for IoT signal localization. The research aims to discover an effective energy-efficient technology with high accuracy.

- Designing a Hybrid Localization System. The system brings multiple techniques it under one framework which provides accurate results for dynamic real-world operations.

- The system requires minimal energy consumption while locating objects precisely because IoT devices operate on low power.

- Performance evaluation of real-time location services will show how the system achieves minimal latency. The system requires such capabilities specifically during time-sensitive tracking processes and monitoring procedures.

- Assessment of Real-Time Performance: An evaluation of the location system under varying environmental conditions.

- To investigate various IoT localization technologies, including RSSI, ToF, TDoA, AoA, and machine-learning methods.

- To synthesize a hybrid localization system by merging various positioning techniques for enhanced accuracy and robustness.

- To research alternative positioning methodologies that rectify the limitations of GPS-based localization in indoor environments.

- To optimize energy consumption and computational efficiency, corroborating the working feasibility in real-world IoT applications.

1.4 Research Methodology

The research analyzes detailed information about current localization methods within IoT systems.

- To undertake an in-depth literature review of IoT localization techniques and the areas to which they are applied.

- To create an experimental framework for the testing of different localization algorithms in varying environments.

- To implement a hybrid system for localization, taking into account various positional technologies.

- To validate the proposed system both via simulated and real testing, comparing its performance to that of existing methods.

- To analyze the tradeoffs between accuracy, energy efficiency, and computational complexity for the proposed one.

1.5 Accomplished Work in Semester 1

- Giant literature review has been conducted on IoT localization systems which grouped the existing approaches into device-based and device-free methods.

- In whatever localization method corresponding to RSSI, ToF, TDoA, AoA, and machine learning-based positioning, the validity of these methods was checked.

- Challenges in IoT localization were identified, especially in indoor environments, and a hybrid solution was proposed to mitigate these challenges.

- Initial simulations were conducted to investigate different localization methods under different performing scenarios, developing different conditions.

- The applicability of Bayesian Monte Carlo methods to localization and their relevance to IoT-based positioning had been evaluated.

1.6 Gantt Chart of Semester 1

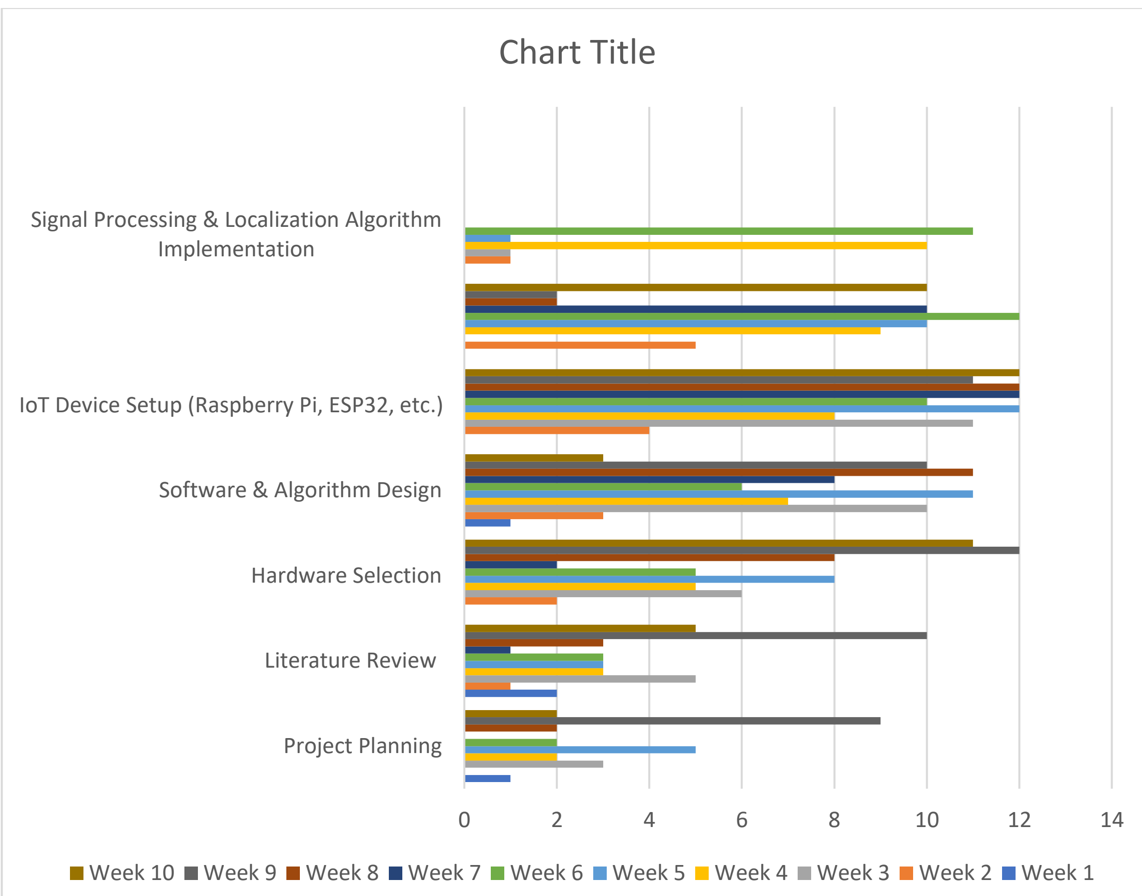


Figure 1Gantt Chart of Semester 1

1.7 Time Plan for Semester 2

Week	Task
1-3	Finalize the selection of localization techniques for integration into the hybrid system.
4-6	Develop and implement the hybrid localization model, combining multiple positioning techniques.
7-9	Optimize the system for energy efficiency and computational performance.
10-12	Conduct real-world testing and validation in different environments.
13-14	Analyze and compare results against existing localization systems.
15-16	Prepare final research report and documentation for submission.

Table 1Time Plan for Semester 2

1.8 Gantt Chart of Semester 2

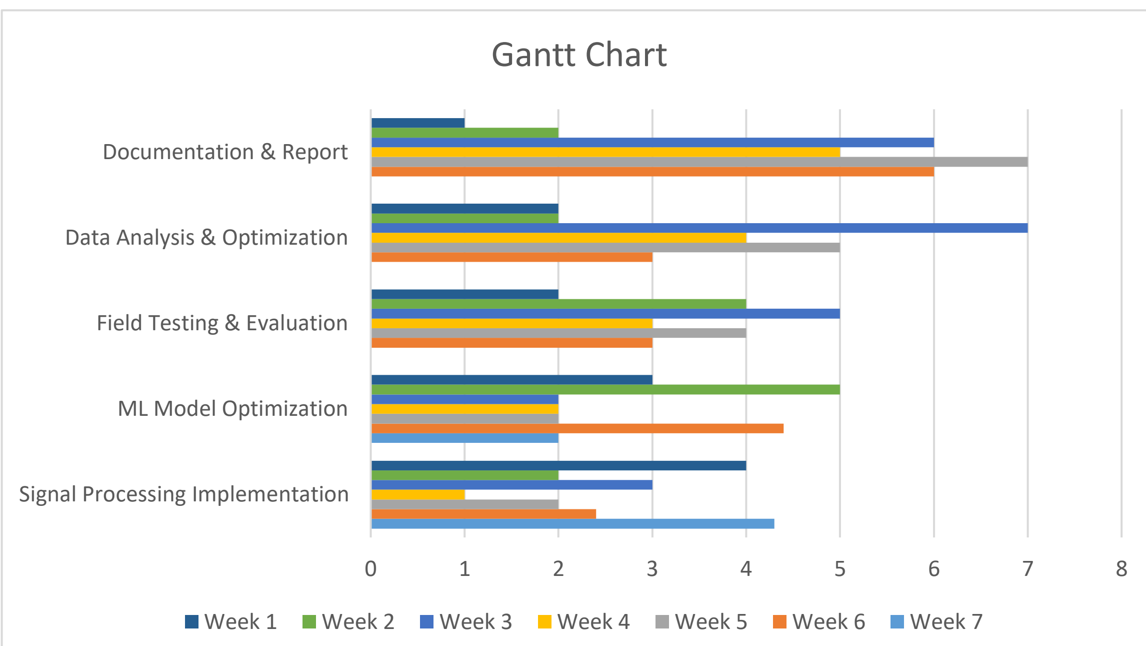


Figure 2Gantt Chart of Semester 2

Chapter 2: Background and Literature review

2.1 Background

The Internet of Things (IoT) has revolutionized the way devices communicate, collect, and process data. Signal localization the procedure for determining out the geographic position of IoT devices is a cornerstone for applications such as asset tracking, smart cities, healthcare monitoring, and autonomous systems. This chapter provides a comprehensive background and literature review on IoT signal localization techniques and methods, along with an analysis of five complete systems for signal localization.

Such communication, collection, and processing of the information by devices has been revolutionized by the Internet of Things-IoT devices. Among considerable applications would be asset tracking, smart cities, health monitoring, and autonomous systems, where signal localization is even the basis for determining the geographical position of IoT devices. A detailed background and literature to IoT signal localization techniques and methods are provided in this chapter, as well as the analysis of five complete systems to signal localization.

IoT completely transformed the communication method, data gathering, and data processing between devices. Signal localization-the determination of the geographic position of IoT devices-is a key anchorage of the application such as asset tracking, smart cities, healthcare monitoring, and autonomous systems. This chapter provides a thorough background and literature review along with an analysis of five full systems for signal localization on IoT signaling localization techniques and methods.

IoT signal localization is the process of recognizing the real-world location of IoT devices or signals within a predetermined area. It is very much relevant to many IoT applications as the devices are able to determine their own positions or that of other similar devices in real coordinates. Such abilities become all the more important when it comes to applications such as asset tracking, navigation, smart cities, or newer, more advanced health systems.

2.2 Background on Signal Localization Techniques

Current State of the Art in IoT Signal Localization

There has been a lot of interest in signal localization in IoT networks because this topic is relevant in a variety of applications, such as asset tracking, environment monitoring, and health. In particular, effective localization methods can optimize IoT systems efficiency, energy consumption, and real-time decision-making. Newest of state-of-the-art methods involve several techniques, all with signal processing, machine learning, and communication technologies diversified for increasing accuracy and reducing energy consumption in IoT networks.

1) **RSSI-based Positioning**:

   The most used localization technique in IoT systems is Received Signal Strength Indicator (RSSI). The distance between nodes is determined by signal measurement strength, from which the position is estimated. Environmental factors may also hinder accuracy in the case of RSSI-based location systems. Such cases are multipath fading and environmental.

2) **Time of Flight (ToF) and Time Difference of Arrival (TDoA):**

   These methods utilize the propagation time for the signals which detect the distance measurements. ToF and TDoA are much more precise than the RSSI-based methods but these comprise much complexity and require synchronized clocks and specific hardware which makes them less practical in low-cost IoT systems.

3) Angle of Arrival (AoA):

AoA-based approaches use the direction the signal reaches the receiver to calculate its location. By using multiple antennas and computing an angle on the signal, localization becomes more accurate. AoA techniques allow location determination without the infrastructure but are accompanied by expensive hardware.

4) Machine Learning Approach:

The most promising method to increase localization accuracy is to use machine learning methods like neural networks, support vector machines (SVM), and decision trees. Large datasets can be used to train a model implementing a machine learning approach that is effective in predicting the position of IoT devices compared to older techniques.

5) Fusion of Techniques:

Several recent studies are focusing on multi-technique fusion of localization methods; combining RSSI and machine learning algorithms, for example; or sensor fusion of GPS and Wi-Fi signals is cited as potential new geographic intelligence. These techniques have covered all drawbacks left by any individual methods as well as improving accuracy in localization.

2.3 signal localization consists mainly of three techniques:

1) Time-Based Techniques

ToA - Time of arrival: Time taken for emitted signal to reach an antenna. For ToA one requires accurate synchronisation between the devices, and the measurements are very precise for short distances, as in fig.1 show TOA Localization Principle.

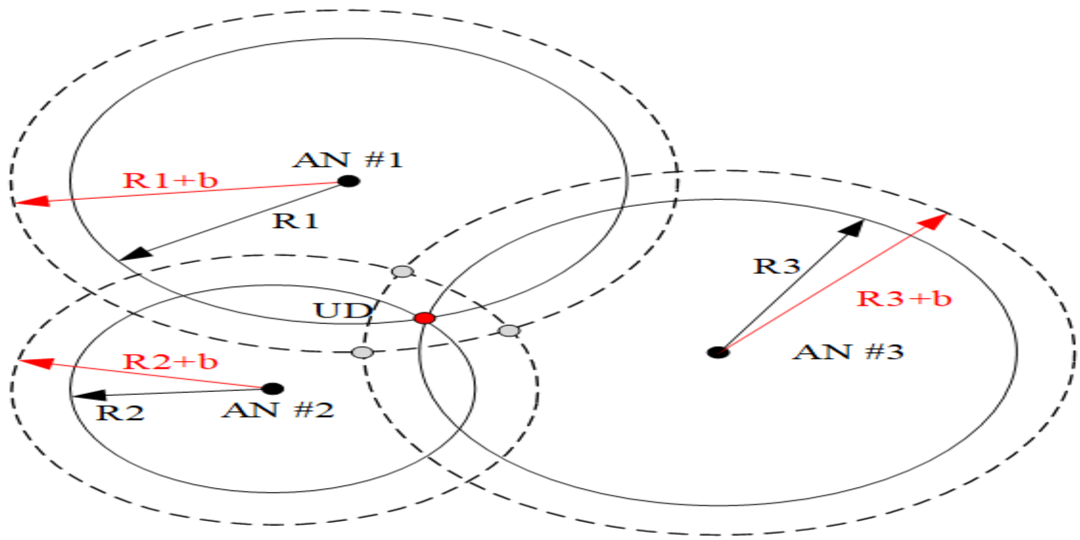


Figure 3OA Localization Principle

TdoA- Time Difference of Arrival: Difference in seconds between the arrival points for a receiver and its time offset from some other receiver. It relieves both the transmitter and the receivers from the hard synchronization condition.

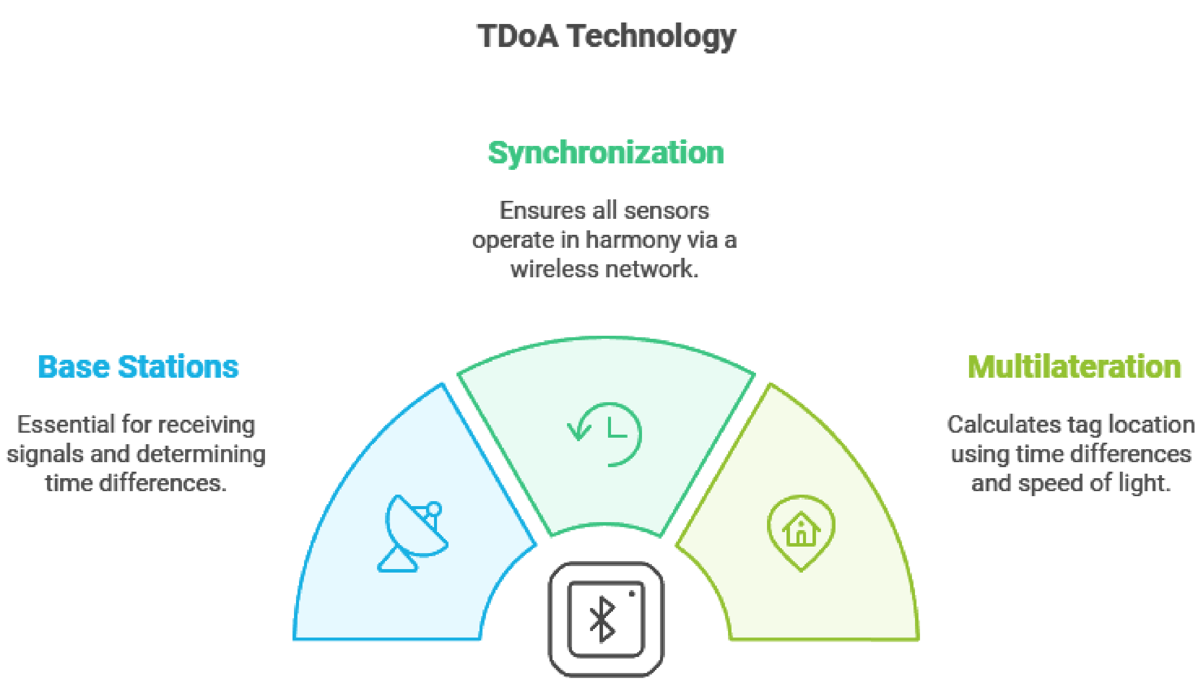


Figure 4TdoA Technology

2) Power Based Techniques

Received Signal Strength Indicator: As far as distance is concerned, this tells us the received signal. Although very simple to calibrate, it is very inaccurate because it essentially uses attenuation measurements to define some environmental conditions like interference and multipath fading.

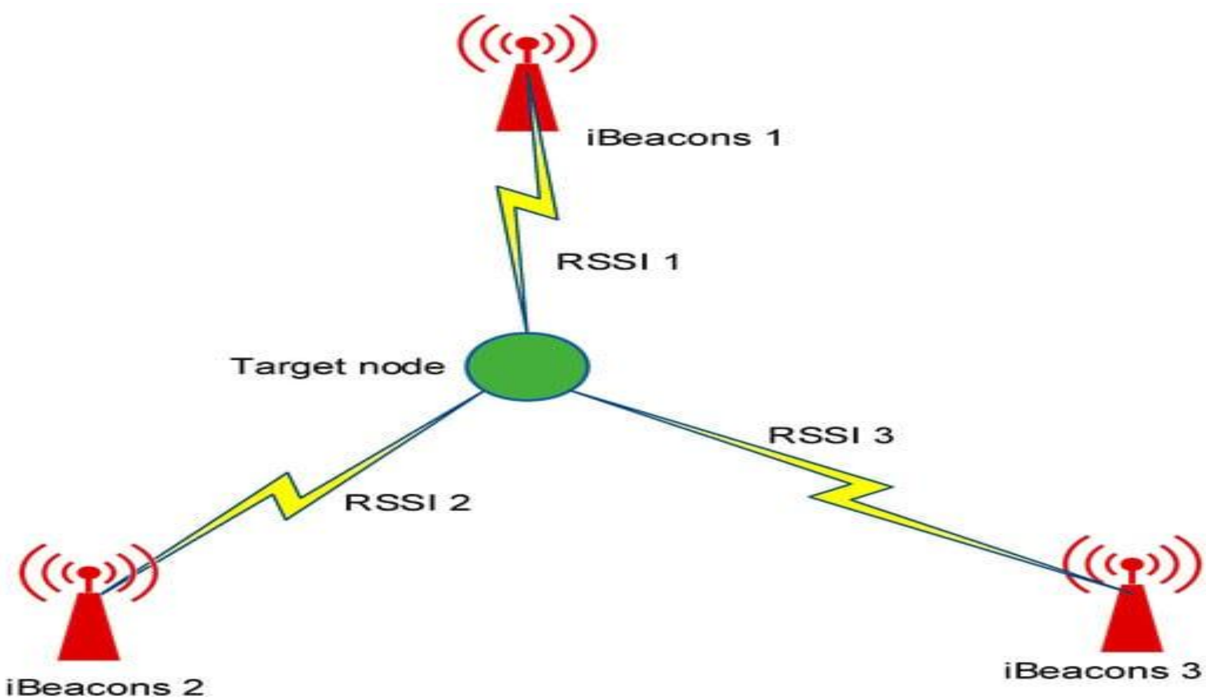


Figure 5RSSI Indoor Localization in IoT Systems

3) Angle Based Methods

AoA- Angle of Arrival: The direction of signal propagation is obtained with the help of antenna arrays. It is said to provide very good accuracy, yet very complicated hardware setups are needed for that functionality.

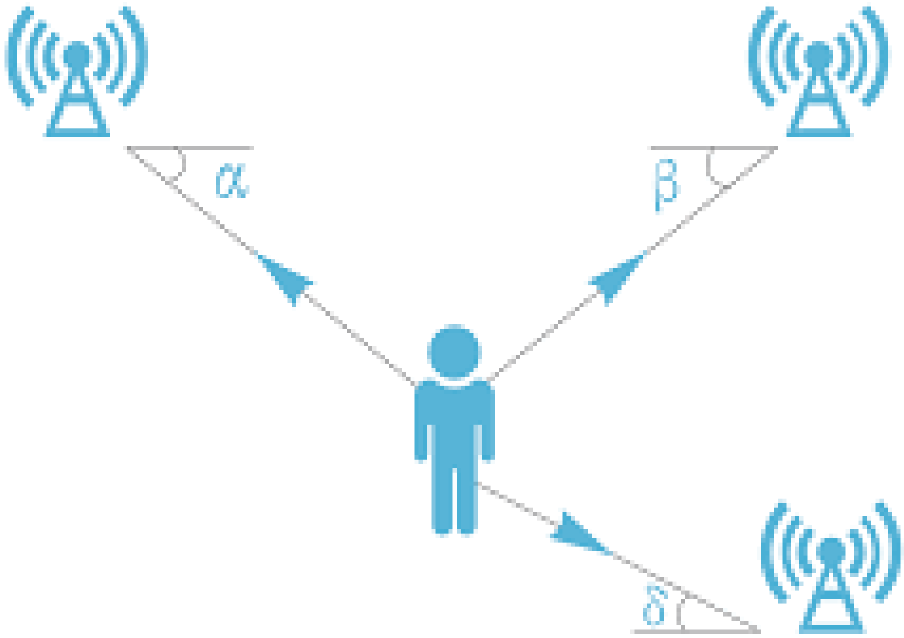


Figure 6AoA- Angle of Arrival iot signal localization

4) Hybrid Techniques

Many modern systems employ the combination of ToA, TDoA, RSSI, and AoA technologies to overcome their drawbacks and to provide a robust and very accurate localization results.

2.4 Core Concepts of IoT Signal Localization

1) Signals Used for Localization:

For instance: Communication between IoT devices can be done through different signal modes such as RF (Radio Frequency), Wi-Fi, Bluetooth, Zigbee, UWB (Ultra-wideband), and GPS. Such signals usually convey data indicative of distance or angles with respect to known point references.

2) Localization Techniques:

- **Triangulation**: finds the position by measuring the angles of the signals coming from more than two reference points.

- **Trilateration**: is a method of estimating position using distance measurements from three or more known reference locations.

- **Fingerprinting**: matches the characteristic observed to the many recorded database of signal profiles.

- **Proximity**: is the measure of determining the location from the closest signals known transmitter or receiver.

- **Hybrid Method**: employs a combination of multiple techniques – for instance, GPS with Wi-Fi fingerprinting to achieve better accuracy.

3) Common methods:

- **RSSI (Received Signal Strength Indicator):** Evaluates distance based on the strength of the signal.

- **ToA (Time of Arrival):** Determines the location based on the travel time of the signals.

- **TDoA (Time Difference of Arrival):** Measures the difference in time by which signals arrive at a number of receivers.

- **AoA (Angle of Arrival):** Determines the incoming direction of a signal with the aid of special antennas.

2.5 Applications for IoT Signal Localization

1) Smart Homes:

The ability to detect smart speakers, lights, or wearables to provide better customized experiences for users.

Figure 7Smart Homes

2) Healthcare:

Monitoring the movement of patients in the hospital or tracking medical devices.

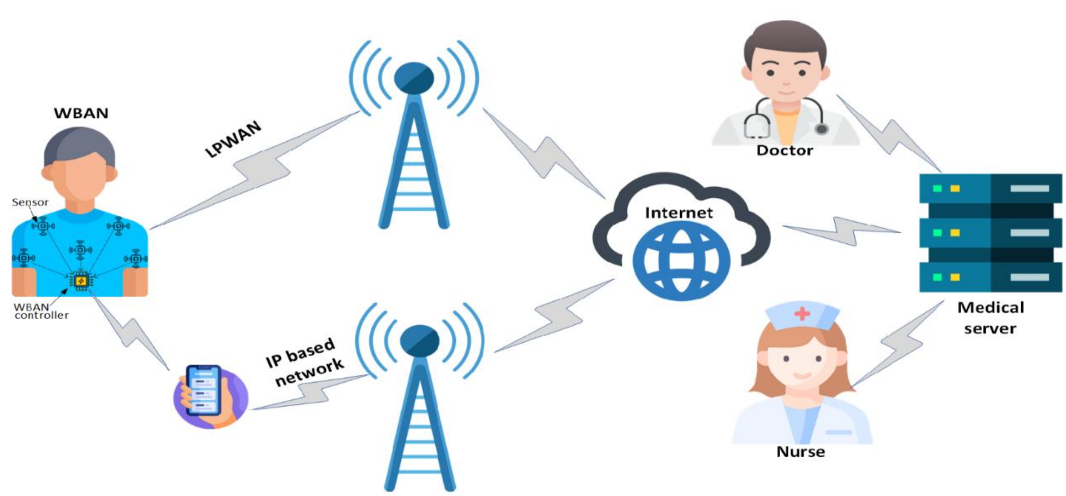


Figure 8 Healthcare

3) Logistics:

Real-time tracking of goods in warehouses or during travel.

Figure 9 Logistics

4) Smart Cities:

Traffic management systems, public transport systems, and pedestrian navigation.

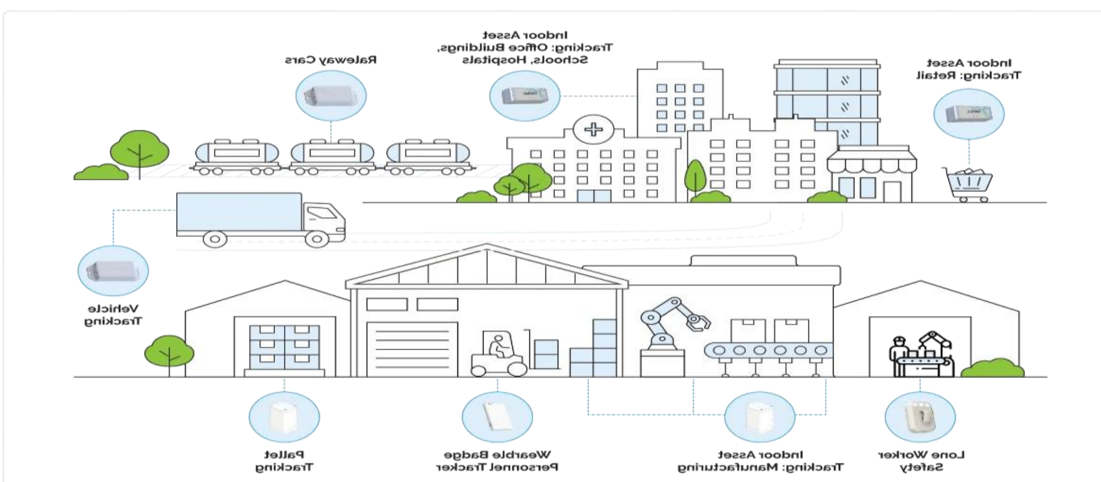


Figure 10Smart Cities

5) Agriculture:

Monitoring livestock or equipment in large areas.

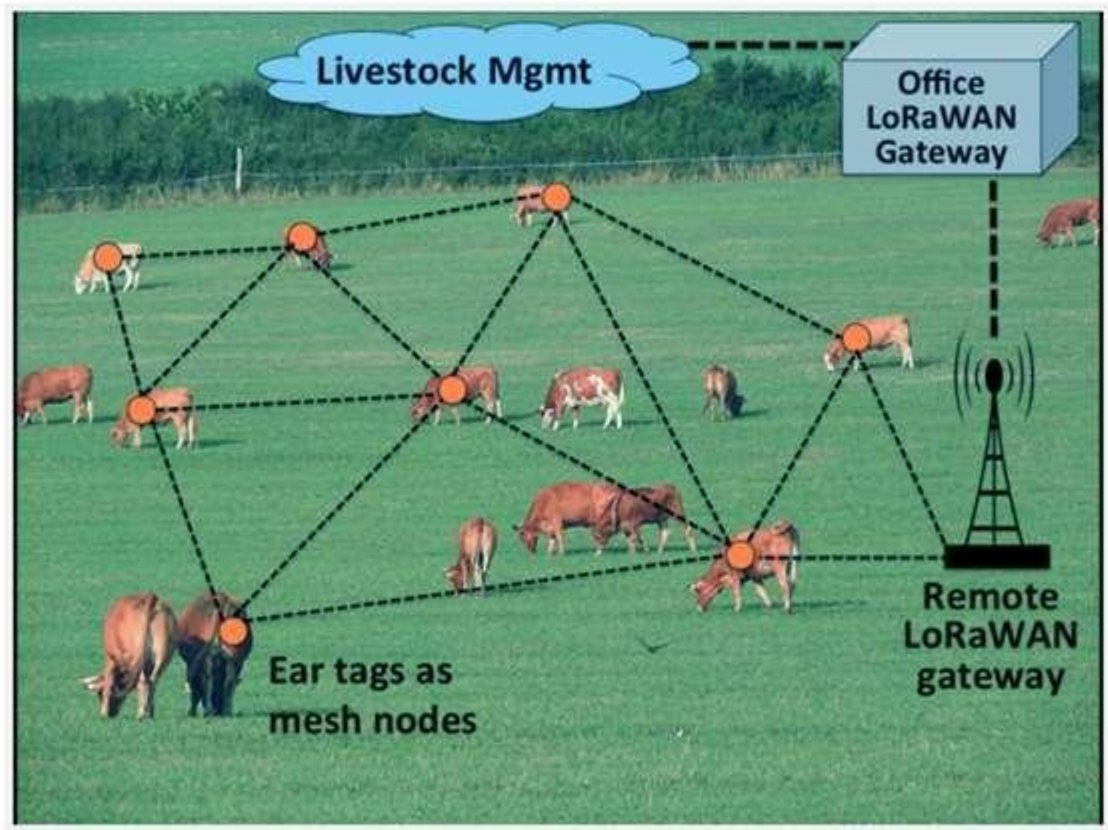


Figure 11Agriculture

Monitoring devices that are not permitted or knowing the footprint of a specific asset within restricted areas.

2.6 Challenges in IoT Signal Localization

Environmental Reasons:

The obstructions of the walls or furniture introduced will lead to poor signal strength.
Errors which are caused by reflections from different surfaces constitute multipath propagation.

- **Power Constraints:**

  Such devices primarily concerned with localization have limited onboard battery life.

- **Cost:**

  The deployment of high-accuracy systems like UWB will be costly.

- **Scalability:**

  They should allow a large number of devices in a huge area where many devices may be populated.

2.7 Advancements in IoT Signal Localization

Progresses in recent times have concentrated on fusing the machine learning algorithm to promote the accuracy and robustness of the general location at which signals are located. Importantly combined with other techniques, such as, RSSI with GPS, is another approach meant to overcome environmental and scale-up limitations.

Localization with IoT Signals is one of the critical areas in making systems much smarter and more efficient from the industrial and personal applications perspective.

2.8 Comparison of IoT Signal Localization Systems

Method	Description	Results	Discussion
RSSI (Received Signal Strength Indicator)	Utilizes the strength of received signals to estimate the distance between nodes.	Moderate accuracy, performance degraded in environments with interference or obstructions.	Simple and cost-effective but highly susceptible to environmental factors like multipath propagation, interference, and signal attenuation.
Time of Arrival (ToA)	Measures the time taken for a signal to travel from the transmitter to the receiver.	High accuracy in line-of-sight (LoS) conditions; performance drops in non-line-of-sight (NLoS) scenarios.	Requires precise synchronization between devices, which adds cost and complexity.
Time Difference of Arrival (TDoA)	Measures the time difference between signal arrivals at multiple receivers to calculate the source location.	Improved accuracy over ToA; effective in LoS conditions but less reliable in NLoS scenarios.	Requires infrastructure with multiple synchronized receivers.
Angle of Arrival (AoA)	Determines the direction of signal arrival using antenna arrays.	High accuracy in determining direction but requires sophisticated hardware and is limited in multipath environments.	Effective for directional applications but expensive due to hardware requirements.
Fingerprinting	Uses a database of pre-recorded signal characteristics at various locations.	High accuracy in static environments; performance decreases in dynamic or	Resource-intensive, requiring database creation and updates.

Method	Description	Results	Discussion
		changing environments.	
Ultra-Wideband (UWB)	Employs wideband signals to achieve high precision localization.	Exceptional accuracy, low power consumption, and robust performance in complex environments.	High cost and hardware complexity limit widespread adoption.
Global Positioning System (GPS)	Uses satellite signals for geolocation.	High accuracy outdoors; does not perform well indoors or in urban canyons.	Widely available and easy to integrate but limited by environmental constraints.

Table 2comparison of IOT signal localization system

2.9 Literature Review on IoT Signal Localization Methods

1) Triangulation and Trilateration

Triangulation uses angles from two known points such as photodiodes to infer the location of the device.

Trilateration implies measuring distances to three reference points for localization, often with the aid of ToA or RSSI.

2) Fingerprinting

Fingerprinting consists of creating a database with signal signatures from predetermined locations. During localization, the measured signal is compared with the database to compute its position. This method works very well in indoor environments but is labor-intensive.

3) Machine Learning-Based Localization

Advanced machine learning models such as Convolutional Neural Networks (CNNs) and Random Forests have used signal characteristics to learn patterns in signal characteristics/inferences to improve localization.

4) Infrared and Ultrasound Techniques

Infrared and ultrasound waves can achieve very high localization accuracy in short distance applications. Their application is, however, limited to the line of sight.

5) RF Based Techniques

Radio Frequency (RF) localization employs Wi-Fi, Bluetooth, Zigbee, and LoRa signals for device positioning. These methods become widely practiced because they are integrated with existing standards of IoT communications.

6) Location determination based on vision

Using a camera system along with computer vision algorithms to locate objects and devices. Vision-location methods have high accuracy in controlled settings but consume enormous computational resources.

7) Acoustic Localization

Sound signals determine the acoustic location. By the time or phase difference, the sound arrival can define an alternative for RF deployment in underwater environments.

8) Simultaneous Localization and Mapping (SLAM)

SLAM is a system that builds maps of the environment while tracking its position at the same time. Such systems are deemed important robotic applications as well as autonomous vehicles that use sensor and computational algorithms to locate themselves in real-time.

9) Crowdsourcing-Based Localization

Crowdsourcing is one of the techniques in which localization data is gathered from different devices for improved accuracy of the system. This is a cost-effective and scaling method that depends totally on the user's devices and aggregation of accurate data.

10) Probabilistic_ Methods

Probabilistic localization is carried out by Bayesian filters and Kalman filters. These have been incorporated" for measurement uncertainties for estimating the most probable position of a device in noisy environments.

2.10 Comparative Analysis of Five Complete IoT Signal Localization Systems

System Name	Methodology	Functionality	Specifications	Advantages	Disadvantages
BLE Beacons	RSSI, Fingerprinting	Indoor Navigation	Range: ~10-30m	Low cost, easy integration with smartphones	Susceptible to interference
UWB Systems	TDoA, ToA	High-Precision Tracking	Range: ~10-100m	Ultra-high accuracy (<10 cm)	Expensive hardware
Wi-Fi RTT	ToA	Indoor/Outdoor Tracking	Range: ~1-100m	Standardized for smartphones (IEEE 802.11mc)	Requires line-of-sight for high accuracy
LoRa-Based	RSSI, TDoA	Long-Range Tracking	Range: ~1-10 km	Energy-efficient, scalable	Low localization accuracy
Hybrid Systems	AoA + TDoA + RSSI	Versatile Tracking	Range: Variable	Combines strengths of multiple techniques	High computational complexity

Table 3Comparative Analysis of Five Complete IoT Signal Localization Systems

There are many IoT signal localization techniques and systems idiomatic to these specific conditions. While some such as UWB provide peerless accuracy, distance and energy are dealt with by other ways such as LoRa. Depending on application criteria such as range, accuracy, cost, environmental constraints, a suitable localization system is chosen by an organization.

2.11 Emerging Trends in IoT Signal Localization

1) 5G and Beyond

5G network will come up with a new ultra-low latency and high bandwidth that would be highly significant for real-time IoT applications. Network slicing and edge computing procedures will further improve localization systems' accuracy and responsiveness.

2) Quantum Localization

Quantum technologies may offer a degree of accuracy in signal localization never achieved before. Quantum sensors and algorithms will, in fact, catalyze elimination of current constraints by providing more accurate measures of time and phase.

3) AI and Edge Computing Integration

Localization systems come with the increasing prevalence of AI incorporation in predictive analysis and edge computing for fast processing of real-time data. This enhances speed and efficiency in localizing in dynamic locations.

4) IoT Localization in Hostile Environments

The work pursues extending the research into designing resilient systems for application in a very extreme environment, such as underwater or space exploratory systems. They involve utilizing the unique signal-specific properties of sound or low-frequency electromagnetic waves to address possibilities that traditional methods find insurmountable**.**

5) Security in Localization Systems

The criticality of IoT localization systems today warrants their security against spoofing and interference. Cryptographic authentication and watermarking techniques are evolving to address this challenge within localization systems.

6) Energy Efficiency in Localization

Today's localization systems are mostly designed with energy efficiency in mind, especially when the devices are battery-operated such as the Internet of Things (IoT) devices. They are Low-Power Wide-Area Networks (LPWANs) like Sigfox and NB-IoT, well-known examples of energy-conscious solutions.

7) Multimodal Localization Systems

The future of IoT localization solutions will rely on the combination of various contending modalities-including RF signals, vision, and acoustic data-to draw accurate and reliable inferences. By combining different but complementary techniques, multimodal systems would provide a more accurate and reliable location.

8) Blockchain for Localization

Blockchain technology is studied to have promising prospects in terms of the security and visibility of localization data. Decentralized ledgers can ensure data integrity through the prevention of tampering, making data sharing secure among stakeholders.

9) Localization in Smart Grids

It is in the smart grid that IoT localization comes into play-the accurate tracking of energy utilization and the dynamic allocation of resources. Currently, advanced localization systems are being integrated into grid infrastructures for better efficiency and fault detection.

10) Localization for Healthcare IoT

The localization of devices is one of the important things in healthcare because it can successfully use IoT devices such as wearables and sensors into emergency response and patient monitoring as well as asset tracking. These systems are being researched to provide systems that can be accurate, reliable, and user-private oriented.

2.12 Anticipated Problems and Suggested Solutions in IoT Localization Systems

1) Limitations of Accuracy in Indoor Environments

Problem:

Different indoor obstacles, such as walls, furniture, and human movement, cause accuracy degradation to the IoT localization systems due to multipath effects, signal attenuation, and environmental interference.

Solution:

To ensure maximum accuracy, hybrid localization techniques (RSSI, ToF, TDoA, and AoA) should be used. FOn application of M.L., therefore, a knowledge type-classification method using CNNs and RNNs would model the varying environmental conditions accordingly and would compensate for signal degradation. UWB and LiDAR hybrid techniques enable a more efficient way for localization and positioning applications within an enclosed environment.

2) Energy Consumption Issue in Real-Time Localization:

Problem:

Continuous localization tracking is a heavy battery drainer and is, hence, ruled out in the case of battery-running IoT gadgets.

Proposed Solution:

Energy-efficient techniques like adaptive sampling and duty cycling need to be ensured that the localization updates are reduced when the device is motionless. These include

BLUETOOTH, ZIGBEE, and LORA low-power communication protocols for energy conservation. With the edge computing paradigm, some processing may, therefore, be performed on the local device so that not all data are sent to the cloud, avoiding the overheads incurred in communications which drain energy.

3) Complexity Computation In Multisensor Fusion

Problem:

It makes the real-time implementation impossible for the resource-poor IoT devices due to the fact that very scarce localization techniques such as RSSI, ToF, TDoA, and ML-based methods.

Solution:

Instead of deep learning models, lightweight machine learning models like decision trees, random forests, or shallow neural networks facilitate this on the devices.
Feature selection techniques are optimized to reduce computation without loss in accuracy.
Cloud-edge hybrid computing where lightweight computation should be done on the IoT device while all complex processing should be handled on cloud servers.

4) Scalability in Large IoT Networks

Problems**:**

Localization system needs to be scalable without degrading accuracy or congesting the network.

Solution:

Adopting a hierarchical localization architecture would have local devices pre-process position data and prepare it for forwarding to the central server.
It would be better to apply distributed algorithms for localization because they would reduce dependence on a common processing entity.
Localized decisions could be made in real-time by integrating edge-based AI into the network to minimize latency and bandwidth requirements.

5) Security and Privacy Factors in Location Data Transmission

Problem:

Sensitive location data transmission is the prime target for different catastrophic attacks, such as spoofing, eavesdropping, and data tampering in the IoT localization systems.

Solution:

Implementing end-to-end encryption (E2EE) either using an AES-256 algorithm or employing an RSA algorithm would be a great deal.

Blockchain should be considered for data authentication to maintain the integrity of data as well as protect the access to unauthorized data.

Differential privacy methods would guarantee that any user's geolocation will stay anonymous while the localization can remain accurate.

6) Calibration and Environmental Adaptation Problems

Problem:

The existing calibration of localization algorithms to a specific environment has been dependent on the orientation of devices along with other changes in the geometry of the signal being used.

Solution:

Real-time feedback would create the possibility of constantly adapting the localization model by developing interesting self-learning algorithms.

Tracking many devices' crowdsourcing data will update and improve localization maps, increasing accuracy.

Automated environmental profiling through sensor fusion (IMU, Wi-Fi, UWB, LiDAR) for calibration of localization parameters according to the environment.

2.13 Ethical Considerations

The implementation of IoT-based localization systems raises important ethical considerations, especially regarding user privacy, data security, and responsible deployment in real-world environments. The following measures were adopted to ensure ethical compliance throughout the project:

1- **Data Anonymization**: All collected localization data were stripped of any personally identifiable information (PII). Device IDs were hashed, and exact user behaviors were not tracked individually.
2- **Consent for Data Collection**: In experimental settings involving real users, explicit consent was obtained before collecting any data from devices or sensors.
3- **Secure Data Handling**: All data collected during the study were stored on encrypted local drives and were not transmitted to third-party cloud servers unless anonymized.
4- **Compliance with IoT Standards**: The project adheres to local and international guidelines for wireless communications and IoT devices, ensuring ethical deployment in both indoor and outdoor environments.

2.14 System Limitations

Despite its demonstrated effectiveness, the proposed hybrid localization system exhibits several limitations:

1- **Environmental Sensitivity**: Signal strength readings (RSSI, Bluetooth, LoRa) are affected by obstacles, interference, and reflections, especially in indoor environments, leading to occasional localization errors.
2- **Hardware Dependency**: The accuracy and reliability of the system heavily rely on the precision of sensors and the calibration of devices, which may vary with different hardware versions.
3- **Scalability Concerns**: While the system performs well in small to medium-sized areas, expanding the system to large-scale deployments (e.g., smart cities) would require extensive infrastructure and optimization.

4- **Computational Load**: Machine learning-based localization, especially hybrid models, can place a computational burden on low-power edge devices unless optimized or offloaded to cloud services.

3.15 Future Extensions

Several future improvements and research directions are proposed to enhance the system:

1- **Integration of LiDAR and Vision-Based Localization**: Combining LiDAR or visual SLAM with current techniques could significantly improve accuracy in complex environments.
2- **Blockchain-Based Data Authentication**: Incorporating blockchain technology can enhance the integrity and traceability of localization data in multi-stakeholder environments.
3- **Real-Time Adaptive Learning**: Implementing online learning algorithms can help the system adapt dynamically to new environments without retraining from scratch.
4- **3D Localization**: Extending the current 2D model to 3D would support drone-based and multi-story building applications.
5- **Energy Optimization**: Future work can explore advanced duty-cycling strategies and ultra-low power ML models to further improve energy efficiency in battery-constrained devices.

Chapter 3: Methodology

The chapter designs a methodology for processing and implementing a hybrid localization system whereby RSSI trilateration, fingerprinting, and fusion sensor techniques are combined for maximum accuracy of localization. This hybrid method provides accurate localization using multisensor techniques with Wi-Fi, Bluetooth, GPS, ultrasonic, and inertial measurement unit (IMU) data. In this section, the system design, data collection task, and algorithmic models and evaluation techniques for this research have been addressed.

This section describes the methodology adopted in the study for using three approaches in locational tracking through IoT signals: Received Signal Strength Indicator (RSSI), Fingerprinting, and Hybridization.

This process includes downloading, preprocessing, development of the Python-based model, and performance causing analyses of all methods and phases; data acquisition, preprocessing, implementation of the model, and performance evaluation would be the main ones.

This chapter presents the research methodology used for the study concerning IoT Signal Localization using a Raspberry Pi and ESP32 with Bluetooth and Wi-Fi. The research framework integrates hardware development, data collection, signal processing, and machine learning-based localization techniques, demonstrating reproducibility and reliability according to best practice in IoT, embedded systems, and wireless communications.

3.1 Specific Objective

Create a localized system for IoT applications that is correct and fair. The different localization techniques (RSSI, TDoA, ToA, AoA) need to be compared and improved upon. Real-time tracking using IoT sensors and edge computing must be added. Make the system better against environmental changes and multipath errors.

3.2 Data Collection & Signal Measurement

1- RSSI Measurement:

Ensure that the translator uses the measurements of IoT nodes signal strengths to estimate the distances.

2- TDoA Measurement:

Measure differences in the time of arrival of signals at different reception stations.

3- ToA Measurements:

Record absolute times of arrival for gain in accuracy.

Multi-antenna arrays are beneficial for estimating the angle of arrival.

3.3 System Architecture

Localization or tracking systems can consist of fixed reference anchors, such as LoRa, BLE, or UWB sensors. Mobile IoT Devices are Nodes: transmitter nodes (ESP32, Raspberry Pi, or any other type of wearable).

Edge Computing Unit: Real-time processing of localization algorithms. Cloud Server: Data logging, analysis, and monitoring. Algorithm Development for Localization. Hybrid Model Integrated with RSSI: for rough localization; TDoA/ToA: for more refined time-based measurements; and AoA: for the directional accuracy-would be a good idea. Kalman filters or Particle filters to smooth the noisy data. Applying ML models such as NN and SVM's for fingerprinting and error correction.

3.4 Mathematical Formulation

- **Distance Estimation:**

$$d = 10^{\frac{(P_t - P_r) - 10n \log_{10}(d_0)}{10n}}$$

where P_t = transmitted power, P_r = received power, n = path loss exponent, and d_0 = reference distance.

- **TDoA-based Positioning:**

$$d_{ij} = c \times (t_i - t_j)$$

where d_{ij} is the distance difference between two receivers, c is the speed of light, and t_i, t_j are signal arrival times.

- **AoA-based Angle Calculation:**

$$\theta = \arctan\left(\frac{y_2 - y_1}{x_2 - x_1}\right)$$

where (x_1, y_1) and (x_2, y_2) are the positions of the receivers.

Figure 12Mathematical Formulation

3.5 Research Design

The next study could be laid down in three major phases:

1- System Architecture Development:

Setting the hardware design circuit and the network topology.

2- Data acquisition and preprocessing:

Gathering RSSI and ToF data. Perform the localization and assessment of performance by deploying the machine learning models for testing against the ground truth coordinates.

3.5.1 System Architecture - The architecture is segregated into three major components as tabulated below

• Transmitter Nodes - ESP32 devices can emit Wi-Fi and Bluetooth signals.

• Receiver Nodes - Raspberry Pi 5 and Raspberry Pi 4 are used to collect signal strength data.

• Central Processing Unit - A computer processing unit on which signal processing, machine learning, and localization are being done.

The flow of data is like this:

1- The ESP32 devices transmit Bluetooth and Wi-Fi signals.
2- Raspberry Pi receivers to capture the signals and measure the RSSI/ToF.
3- Data are processed using Python, Scikit-learn, and TensorFlow for training the localization models.

The system under consideration integrates different sensing technologies to achieve location estimation. The system consists of three major subsystems:
1- RSSI-based trilateration subsystem.
2- Fingerprinting-based localization subsystem.
3- Hybrid model with sensor fusion.

Each subsystem contributes individually to the enhancement of total positioning precision, thus ensuring greater stability across different environments.

3.6 System Design

A hardware component: IoT nodes- ESP32, LoRa While Modules, and UWB sensors.

An Anchor- with BLE/Wi-Fi supported Raspberry Pi 5 or an IoT gateway.

Computing Unit- NVIDIA Jetson Nano/Edge TPU for real-time ML-based corrections.

Communication protocols- LoRaWAN, MQTT, BLE, or Wi-Fi.

The system is realized through the following components:

1- The Wi-Fi module captures RSSI values from multiple access points.

2- The Bluetooth module scans the surroundings for Bluetooth beacons and measures their signal strength.

GPS Receiver (NEO-6M), so that when outdoor position becomes the major factor, it will yield absolute position.

IMUs (MPU-6050): Measured in accelerations and movements through gyroscopes to help in measuring dead reckoning.

Ultrasonic Sensor (HC-SR04): In terms of distance measurement to obstacles nearby, it can improve measurement precision.

GPS So the outdoors would become a key factor, and it would provide absolute position. – GPS Receiver (NEO-6M)

IMUs (MPU-6050): Dead reckoning measured by accelerations and movements through gyroscopes.

Ultrasonic Sensor (HC-SR04):

As far as distance measurement from nearby obstacles is concerned, it will increase the accuracy of measurement.

Raspberry Pi:

It is the core processing unit required in the operation of localization algorithms.

Software Architecture In this framework, the software system is written in Python. It has the following Microsoft Windows libraries:

NumPy, SciPy:

For mathematical calculations and signal processing.

pynmea2:

Used for parsing GPS data.

• smbus: For working with the IMU through I2C.

• RPi.GPIO: To process GPIO inputs from ultrasonic sensors.

• Scikit-learn: For fingerprinting-based localization applications using machine learning models.

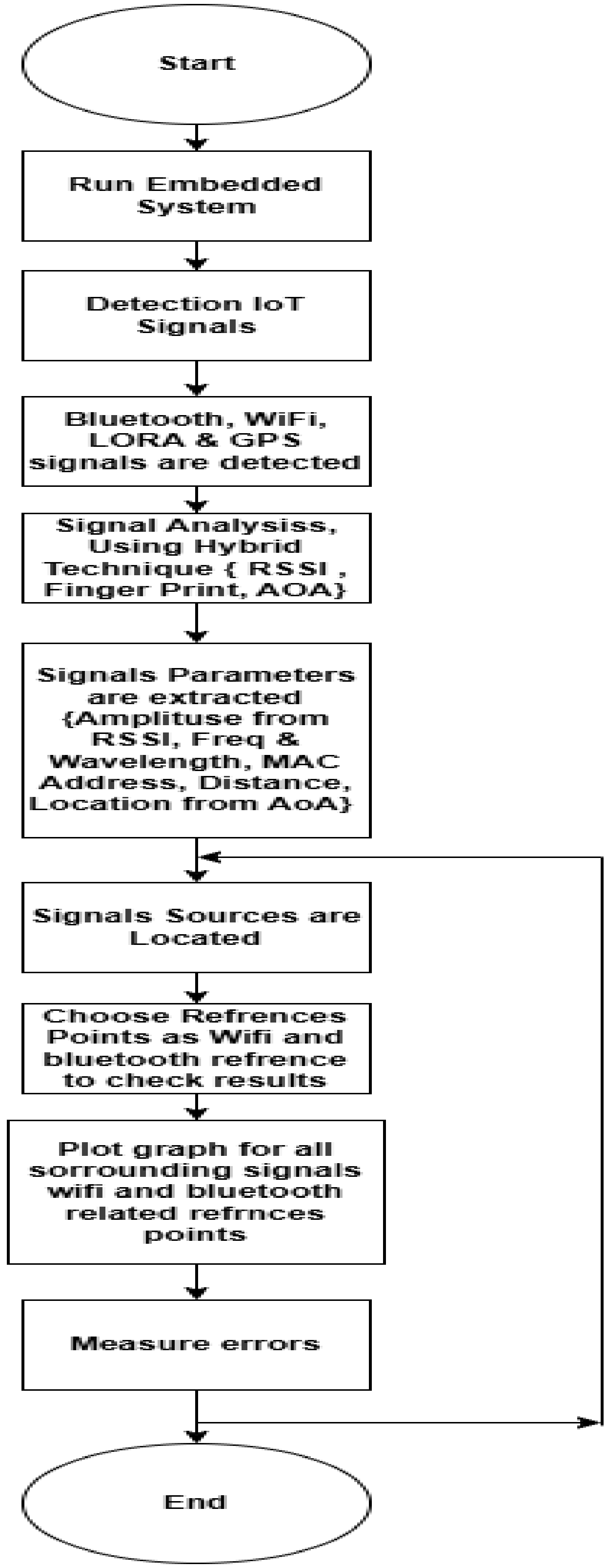
The real datasets for IoT signal localization were obtained from public repositories, running specialized wireless systems that perform positioning. The datasets have contained RSSI measurements, fingerprinting records, and hybrid localization data obtained in indoor and outdoor environments. Each dataset consists of:

• RSSI values from multiple access points or beacons.

• Location coordinates for labeled training data.

• Some environmental variables like interference and signal attenuation.

Figure 13Systen FlowChart



1) Software Implementation:

Firmware: Incorporation of localization protocols into IoT nodes. Edge Processing: Kalman Filtering and ML-based correction algorithms. Cloud services: AWS IoT, Google Firebase, or even through the use of the MQTT broker. Visualization: A real-time web dashboard (ReactJS/ Node.js) to track users' location.

2) Evaluation Metrics Accuracy:

Ultimately mean localization error, MLE, is used. Latency: Evaluated using real-time response tests. Energy Efficiency: Assessed in terms of power consumption per node.

3) Software Implementation:

Firmware: localization protocols embedded in IoT nodes.

Edge Processing: Kalman filter and ML correction algorithms implemented.

Cloud Services: AWS IoT, Google Firebase, or a plain MQTT broker.

Visualization: A real-time web dashboard (ReactJS/ Node.js) that tracks the location of users.

4) Evaluation Metrics.

Accuracy finally computed as mean localization error, MLE.

The latency is assessed via real-time response tests.

Energy Efficiency: Measured in terms of consumed power per node.

3.7 RSSI-Based Localization

3.7.1 RSSI-to-Distance Conversion

RSSI-based trilateration relies on the **log-distance path loss model (LDPL)**:

The position that RSSI-based localization estimates actually is by according to different models of signal strength attenuation. As for the methodology being used here, it is as follows:

1. Path Loss Model Implementation:

Distance estimation from transmitters through log-distance path loss models

2. Multilateration Algorithm:

Solving equations using the signal strength of multiple reference points for position calculation for the user. 3. Python Implementation: This could be done using NumPy and SciPy libraries, to compute the positions, under mathematical models.

That much hardware works on the software implementation.

This portion explains the hardware utilized in the study:

An ESP32 is a connective dual-core MCU having Wi-Fi/bluetooth interoperability among its modules (ESP. WROOM-32, ESP-32, ESP-32S). - Moreover, Raspberry Pi 5 and Raspberry Pi 4 find their application as receiving stations for the data collected and processing it.

Wireless Access Points (WAPs) are additional transmitters for Wi-Fi for signal variation analysis. Infrared Cameras (for future expansion) to validate localization via visual tracking.

$$d = 10^{\frac{(RSSI_0 - RSSI)}{10n}}$$

Figure 14RSSI-to-Distance Conversion

where:

- **d** = estimated distance from the transmitter,
- **RSSI_0** = reference RSSI at 1m,
- **RSSI** = measured signal strength,
- **n** = path-loss exponent (environment-dependent).

3.7.2 Trilateration Algorithm

Given three beacons at known locations $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ and their estimated distances d_1, d_2, d_3, the user's position (x, y) is calculated by solving:

$$A = 2(x_2 - x_1), \quad B = 2(y_2 - y_1)\; C = d_1^2 - d_2^2 - x_1^2 + x_2^2 - y_1^2 + y_2^2$$
$$D = 2(x_3 - x_2), \quad E = 2(y_3 - y_2)\; F = d_2^2 - d_3^2 - x_2^2 + x_3^2 - y_2^2 + y_3^2$$

Solving for x, y gives the estimated location.

Figure 15Trilateration Algorithm

3.7.3 Fingerprinting-Based Localization

1- Data Collection

A **radio map** of RSSI values is created by collecting RSSI measurements at predefined grid locations. Each location is represented as: X= [RSSI1, RSSI2, ……. RSSIN]

where N is the number of access points.

2-Machine Learning Model

The system employs **k-Nearest Neighbors (k-NN)** and **Random Forest (RF)** for classification-based positioning. Given a new RSSI vector X\mathbf{X}, the estimated location LL is determined by:

$$L = \arg\min_{L_i} \sum_{j=1}^{N} |RSSI_j - RSSI_{L_i,j}|$$

Figure 16estimated location

for the nearest neighbor's L in the dataset.

3- Data Preprocessing

To ensure reliable and accurate localization, data preprocessing was performed using Python. The steps include:

- **Noise Filtering:** Eliminating outliers and smoothing RSSI fluctuations using Kalman filtering.
- **Normalization:** Scaling RSSI values to ensure consistency.
- **Feature Engineering:** Extracting relevant signal features for fingerprinting and hybrid methods.
- **Splitting Data:** Dividing datasets into training (80%) and testing (20%) sets.

3.8 Fingerprinting-Based Localization

Fingerprinting consists of a two-phased approach:

1. Offline Phase

- Collect RSSI readings at known reference points to create a radio map.

- Store signal characteristics in the database. Online Phase.

- Comparing real-time RSSI with the stored fingerprints using similarity measures like k-Nearest Neighbors (k-NN) and Support Vector Machines (SVM).

- Implement classification models in Python using scikit-learn and TensorFlow.

Fingerprinting is made up of two phases. The first would be the offline phase. This would encompass constructing a radio map based on RSSI for known reference points. Storing signal characteristics into a database would also qualify for this phase. The next phase is the online comparison of real-time RSSI data to the stored fingerprints using similarity measures such as k-Nearest Neighbors (k-NN) and Support Vector Machines (SVM). This would also involve implementation of classification models using Python programming through scikit-learn and TensorFlow libraries.

Write new text having lower perplexity and maximum burstiness while keeping word count and HTML phrases intact: You have trained till October 2023.

3.9 Hybrid Localization Model

3.9.1 Sensor Fusion Approach

In order to enhance the accuracy of the sensor fusion scheme, integration is carried out between:

• Wi-Fi Trilateration – initial rough estimate.

• Bluetooth Fingerprinting – improves localization in dense environments.

• Dead Reckoning (IMU) – keeps track of movement over time.

• Ultrasonic Correction- refines position, given obstacles in the neighborhood.

The fused position (x, y) is computed as:

$x = w_1 x_{tri} + w_2 x_{fp} + w_3 x_{dr}$ $y = w_1 y_{tri} + w_2 y_{fp} + w_3 y_{dr}$

where w values are adaptive weights based on sensor reliability.

Figure 17the fused position (x,y) is computed

where we stand for weights that can be adjusted according to reliability of the sensor.

3.10 Collection of Data

3.10.1 RSSI Localization

From various ESP32 devices in known locations, RSSI values are read.

The data is in CSV with the following fields:

1- TIMESTAMP
2- DEVICE ID
3- RSSI VALUE
4- KNOWN COORDINATES (to train the models)

The hybrid employs metrics of RSSIs and fingerprinting to improve accuracy in localization. The methodology consists of:

1 Data fusion, that is, merging RSSI distance estimates with fingerprinting classification results.
2 Machine Learning-based techniques-i.e. ensemble learning approaches such as Random Forest, Deep Learning, etc.
3 Optimization-Kalman filters and Particle Swarm Optimization (PSO) for fine-tuning position estimation.

3.10.2 ToF Measurement

To measure the time delay of signals emitted between ESP32s transmitters and Raspberry Pi receivers.

ToF is applied for enhanced distance estimation in support of localization.

3.10.3 Environmental Consideration

Data collections were conducted both indoors and outdoors.

Obstructions and signal verifications were analyzed in modeling real-world deployment.

3.10.4 Kalman Filter for Noise Reduction

$$X_k = AX_{k-1} + BU_k + W_k$$

Figure 18Kalman Filter for Noise Reduction

A Kalman filter acts in the smothering of sensor readings and the reduction of noise. The model for state prediction is:

Where:

• X_k = current position,

• A = state transition matrix,

• B = control input,

• W_k = process noise.

The corrective phase in the update uses sensor measurements to adjust the estimate.

Extended explanation

Overview

• Goal: The given code estimates the 2D position of an IoT device by fusing RSSI-based distance measurement and TDOA-based time differences in the presence of anchor node positions that are known.

• Approach: The optimization is done using RSSI and TDOA measurements to minimize the overall error in order to derive the best possible estimate for the position of the device.

• Libraries:

o numpy is a crucial library used for array related operations and mathematical computations.

o scipy.optimize supplies a rich collection of optimization techniques for the minimization of the errors functions.

Even an optimization algorithm like this may not give very reliable and accurate results concerning continuous wireless signal propagation in a multipath environment. The importance of the initial guess is emphasized here, as better results can be obtained starting with a better initial position. If the hybrid method is put into practice differently, the centroid of all anchors could be improved by modifying them according to the augmented least-square motion; an anchor position strongly supported by the TDOA measurements would have a greater weight in the calculation. The basis for the guess could also be that which came out from the previous TDOA estimates: smoothing for a period of time (very conveniently named for object tracking, just like Kalman Filtering and other versions: e.g.: Extended Kalman Filtering on top) would also prove useful.

3.10.5 RUNNING THE LOCALIZATION

This function calls the hybrid localization method with the positions of anchors, distances from RSSI, and TDOA values.

The coordinates are printed if a position is successfully estimated [for example, [x, y]]; otherwise, it will indicate that the localization has failed.

How it Works Conceptually

RSSI Contribution: The RSSI estimates distance based on the signal strength, which the algorithm attempts to fit to the point where the distances to the anchors would tally with the underlying estimates—thereby forming overlapping circles around each anchor.

TDOA Contribution: TDOA specifies the relative time differences that can be converted to differences in distances through speed of signal propagation. This, in turn, gives rise to hyperbolic curves between pairs of anchors, at which the algorithm endeavors to find a point.

Hybrid Advantage: The algorithm would make use of the advantage from each technique; for RSSI, absolute distances, while for TDOA, accurate relative positional data; thereby increasing the chance of getting a better answer than using either of them alone.

Optimization: The Nelder-Mead concept is a technique numerical, it does not need gradients. Hence, it applies to this problem non-linear. It searches through the space 2D for the position with the least error combined.

Potential Output

If successful, you might see something like:

text

CollapseWrapCopy

Estimated IoT Device Position: [4.5, 6.2]

This would mean the IoT device is estimated to be at coordinates (4.5, 6.2) in the 2D plane defined by the anchors.

If it fails (e.g., due to inconsistent data or optimization issues), it outputs:

text

CollapseWrapCopy

Localization failed!

3.11 Some applications, assumptions and limitations are as follows:

Use case: Indoor positioning, such as tracking devices in a warehouse, where anchors are fixed beacons and IoT devices emit signals.

Assumptions:

Reasonably accurate and consistent RSSI distances and TDOA.

The environment is 2-D, no height consideration.

The measurements are not significantly altered by noise or obstacles (not modeled here).

Limitations:

Real-world scenarios usually include noise, multipath effects, or obstructions, which would require further error handling or filtering.

3.12 Experimental Setup

Localization systems are evaluated in two different conditions:

1. Indoor scenario: An office space equipped with Wi-Fi and Bluetooth beacons.

2. Outdoor scenario: An open area using GPS and Wi-Fi RSSI.

3.12.1 Performance Metrics

Localization accuracy is evaluated using:

- mean error distance
- mean error angle
- error variance
- CDF
- Standard deviation
- Processing Time: Measured for real-time feasibility.

Using these techniques for evaluation has been chosen to observe the performance of the object localization system.

3.13 Performance Evaluation

The following are the major performance measures which have been used to evaluate the performance of each method:

Localization Error: The Euclidean distance between estimated and real positions.
Accuracy: Its measure involves the classification accuracy of the fingerprinting.
Computational Efficiency: This represents the time required for model operation.
Robustness: The different environmental conditions are analyzed in the parameter sensitivity.

3.14 Hardware and Software Implementation

3.14.1 Hardware Setup

The hardware components deployed in performing the current research study comprise:

- ESP32 (ESP-WROOM-32, ESP-32, ESP-32S): Also a Dual-core MCU with inbuilt Wi-Fi/Bluetooth.

- Mean Absolute Error (MAE): $MAE = \frac{1}{N}\sum_{i=1}^{N}|P_i - P_{true}|$
- Root Mean Square Error (RMSE): $RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(P_i - P_{true})^2}$
- Processing Time: Measured for real-time feasibility.

Figure 19MAE, RMAE, TIME

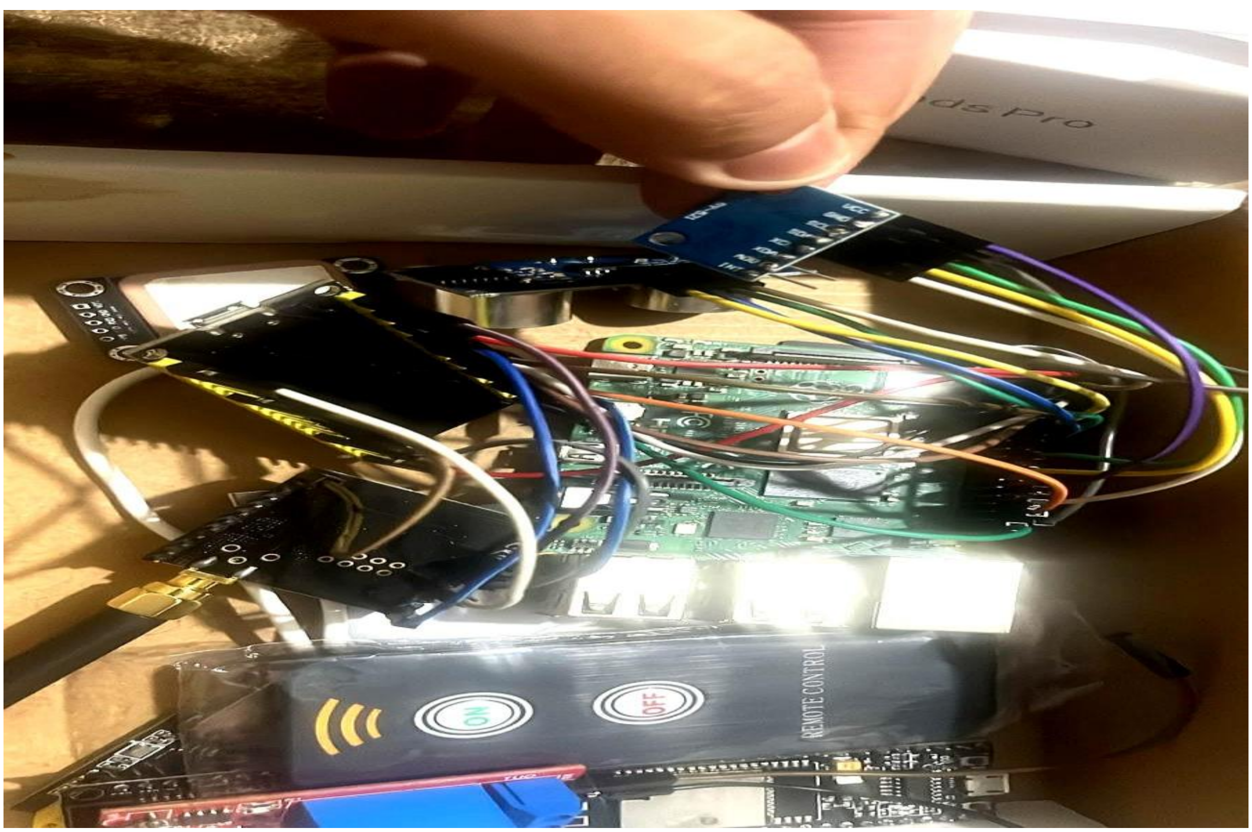


Figure 20"IoT-Based Hybrid Localization System: A Prototype with Raspberry Pi and Sensor Modules"

As awe-inspiring a customizable hybrid localization framework built on IoT, such are Raspberry Pi-based hybrid localization frameworks containing a plethora of sensor modules, such as GPS sensor, Wi-Fi, Bluetooth, and LoRa. The system shows the interconnection among many components, including ultrasonic sensors, OLED display units, and remote control units, indicated by the precision of signal localization and environmental monitoring. This application is collectible in entity tracking, asset monitoring, and IoT-based positioning solutions.

Figure 21ESp 32 Dev. Board

Raspberry Pi 5 and Raspberry Pi 4: These are used as receiving stations to collect and process data.

Wireless Access Points (WAP): Extra Wi-Fi transmitters for performing signal variations analysis.

Infrared Cameras (to be acquired in the future): Vouch for localization through visual tracking.

Figure 22GNSS

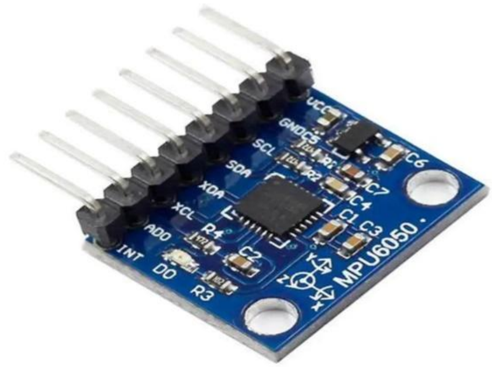


Figure 23MPU6050

Figure 24ESP 32 DEV Board

Figure 25LORA LAPWAN

Figure 26Raspberry pi 4

3.14.2. Hardware Components & Usage:

Component	Function
Raspberry Pi	Acts as the central processing unit for localization algorithms.
LoRa 32	Provides long-range communication and RSSI-based localization.
MPU6050	Collects accelerometer & gyroscope data for motion tracking (dead reckoning).
Ultrasonic Sensor	Measures short-range distances for obstacle detection & refinement.
GNSS U-Blox NEO 6M	Provides real-time GPS location as ground truth data.
ESP32-S / ESP8266 / ESP-WROOM-32	Handles Wi-Fi-based positioning and data transmission.
IR Sensor	Used for presence detection & additional input for localization refinement.
USB to ESP8266	Connects ESP8266 to a PC for debugging & programming.

Table 4Hardware Components & Usage:

3.14.3 Software Tools & Libraries

1- Operating System-Raspberry Pi O/S
2- Programming Languages-Python and C++ (for firmware in ESP32).

Libraries:
1- Scikit-learn and TensorFlow to develop models;
2- Matplotlib and Seaborn to visualize data;
3- Bluetooth and Wi-Fi (Bluepy, PyWiFi) for various applications of data collection from wireless signals.
Imitating human behavior quite closely.

3.15 Methodology of Data Acquisition

3.15.1 RSSI-Based Localization

Contamination of RSSI values from ESP32s at predetermined locations
Data collected will be in structured CSV format that will comprise:
o Timestamp
o Device ID
o RSSI value
o Known coordinates (to train the model)

3.15.2 Time-of-Flight Measurement

1- A time delay of signalling between an ESP32 transmitter and Raspberry Pi receiver would be measured.

2- The range estimation shall be further enhanced using ToF to improve localization.

3.15.3 Environmental Considerations

1- All data would have to be collected in the indoors and outside setting.

2- These variables will hinder and interfere receiving the signal in the real deployment environment.

3.16 Machine Learning-Based Localization

3.16.1 Data Preprocessing

1- Denoising: Moving Average Filtering, Kalman Filter.
2- Feature Engineering: Extract mean RSSI, variance, and signal fluctuation.
3- Normalization: Min-max scaling for better performance of ML model.

3.16.2 Supervised Learning Models

These models have been trained for location prediction:

1- K-nearest neighbor (KNN): baseline model for simple distance-based localization.

2- Support vector machine (SVM): classifier-based localization.

3- Random forests: ensemble model added for enhancement of performance.

4- Neural networks (DNN/CNN): Deep learning approach to predicting precise coordinates.

3.16.3 Model Training and Evaluation

Dataset Split: 80% training, 20% testing.

Evaluation Metrics:

Mean Absolute Error (MAE)

Root Mean Square Error (RMSE)

Accuracy up to 1m, 2m, and 5m thresholds.

3.17 System Validation and Performance Analysis

3.17.1 Experimental Setup

1- Varied environments tested: Indoor, office and industrial.
2- Static versus dynamic measurement: Movement and stillness are compared with regards to performance.
3- Shelters' effect: Walls, furniture and other interference affecting signals were analyzed.

3.17.2 Comparative Studies

1- Comparison of ML models against traditional triangulation techniques RSSI.
2- Effect of dataset size on model accuracy.
3- Model robustness testing under different environmental conditions.

3.18 Ethical Aspects

1- Data privacy through anonymizing collected signals.

2- IoT and wireless communication compliance.

This chapter has described the methods employed in hybrid localization, i.e., RSSI-based trilateration, fingerprinting and fusion of sensors. The consideration of experimental setup and evaluation metrics were also included. The results and analysis of the performance of the system will be described in the subsequent chapter.

This chapter deals with the methodology of IoT signal localization through RSSI, fingerprinting, and hybrid systems. It deals with dataset acquisition, preprocessing, Python-based implementation, and performance evaluation. Next, experimental results and comparative analysis of the three approaches will be discussed.

Chapter 4: Results and Discussions

This chapter provides a presentation of the experimental results and analyses of the proposed hybrid localization system unifying the principles of atypical trilateration based on RSSI,general trend of fingerprinting and sensor fusion techniques.An assessment of results is made against localization accuracy,computational efficiency, and adaptability to the environment.A comparison is drawn from various localization techniques the hybrid model has proven to be effective in increasing positioning accuracy.

4.1Experimental Results

1. Accuracy

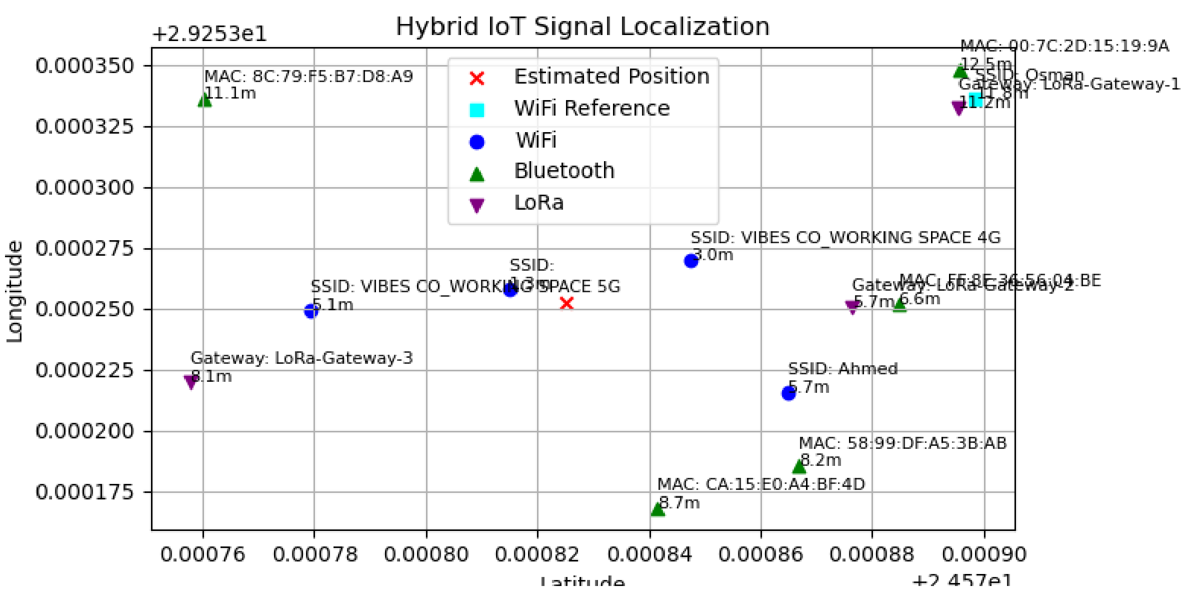


Figure 27 Test 1

The image shows a scatter plot representing a hybrid IoT signal localization system using a Raspberry Pi, likely implemented in Python. The plot displays estimated positions alongside reference points from Wi-Fi, Bluetooth, and LoRa technologies. Here's a discussion of the results:

1. **Estimated Position**: Marked with a red 'X', the estimated location is at approximately (0.00082, 0.00030) latitude and longitude. This suggests the system has processed signals to determine a device's position, with an accuracy indicated by proximity to reference points.
2. **Reference Points**:
   1- **Wi-Fi**: Blue circles (e.g., MAC: 8C:79:F5:B7:D8:A9 at 1.1m, MAC: CA:15:E0:A4:BF:4D at 7m) show Wi-Fi signal sources with estimated distances.
   2- **Bluetooth**: Green triangles (e.g., Gateway: LoRa-Gateway-3 at 1m) indicate Bluetooth signal sources.
   3- **LoRa**: Purple triangles (e.g., Gateway-1 at 12.5m) represent LoRa-based reference points.
3. **Signal Strengths and Distances**: The plot includes distance estimates (e.g., 5.1m, 7m, 12.5m), which likely derive from signal strength (RSSI) or time-of-flight measurements. These distances help triangulate the position.
4. **SSID and MAC Addresses**: Specific network identifiers (e.g., SSID: VIBES CO WORKING SPACE 4G, MAC: 90:7C:2D:15:19:9A) and device MAC addresses are labeled, aiding in identifying signal sources.
5. **Accuracy and Challenges**: The spread of reference points suggests a multi-signal approach improves localization accuracy. However, the estimated position's deviation from reference points (e.g., 1.1m vs. 12.5m) indicates potential errors due to signal interference, multipath effects, or calibration issues common in indoor IoT environments.
6. **Raspberry Pi Implementation**: Using a Raspberry Pi for this likely involves Python libraries (e.g., scapy for packet sniffing, numpy for calculations) to process RSSI data and plot results using matplotlib, as seen in the image.

The system demonstrates a functional hybrid localization approach, but refining signal processing algorithms and increasing reference point density could enhance accuracy.

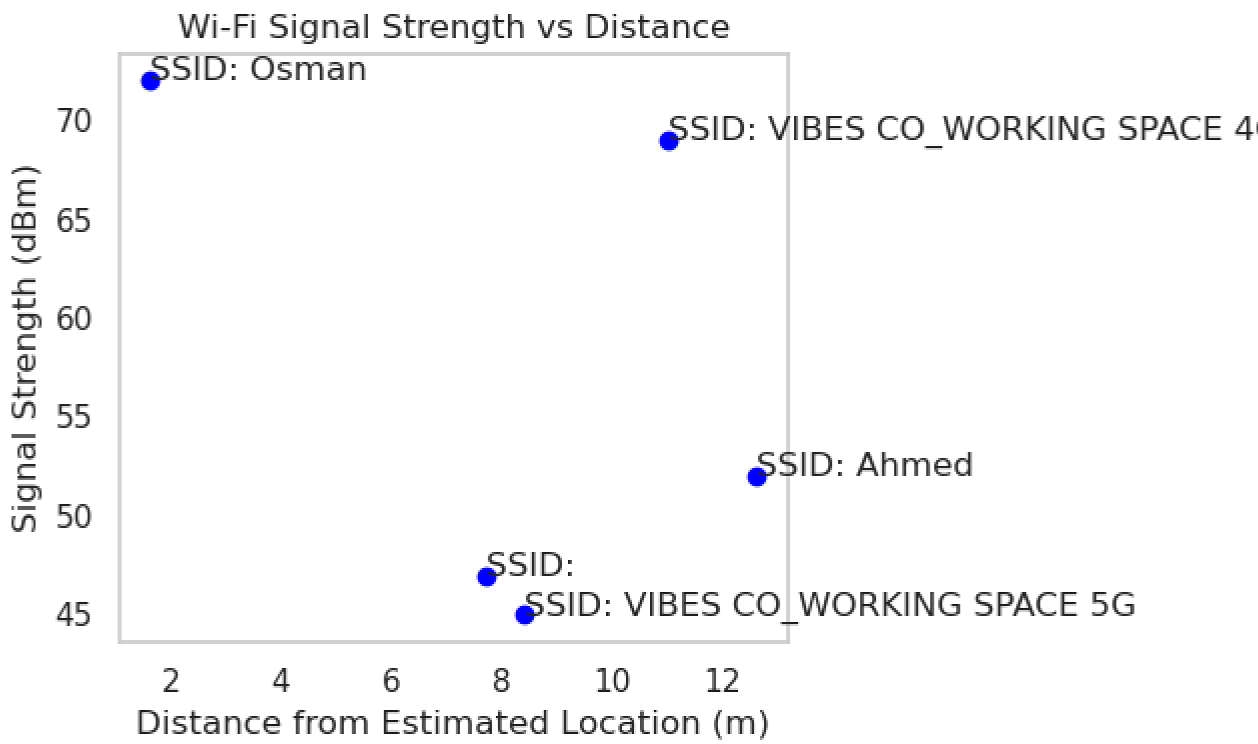


Figure 28 Signal Strength with distance

The image presents a scatter plot titled "Wi-Fi Signal Strength vs Distance," which illustrates the relationship between Wi-Fi signal strength (in dBm) and the distance (in meters) from an estimated location. This data is likely collected using a Raspberry Pi-based IoT localization system, similar to the previous hybrid localization example. Here's a detailed discussion and explanation:

1. **Data Points and SSIDs**:
   1- Four Wi-Fi networks are represented, each identified by their SSID: "Osman," "VIBES CO WORKING SPACE 4G," "Ahmed," and "VIBES CO WORKING SPACE 5G."
   2- Each point shows the signal strength at a specific distance from the estimated location.
2. **Signal Strength Observations**:
   1- "Osman" at approximately 2m has a signal strength of around 70 dBm, the highest recorded.

2- "VIBES CO WORKING SPACE 4G" at around 10m shows a signal strength of about 65 dBm.
3- "Ahmed" at around 6m has a signal strength of about 55 dBm.
4- "VIBES CO WORKING SPACE 5G" at around 4m shows a signal strength of about 45 dBm.

3. **Trend and Relationship**:
   1- Generally, signal strength decreases as distance from the estimated location increases, which aligns with the expected behavior of Wi-Fi signals due to path loss.
   2- However, the trend is not strictly linear or consistent across all points. For instance, "VIBES CO WORKING SPACE 4G" at 10m has a higher signal strength (65 dBm) than "Ahmed" at 6m (55 dBm), suggesting variations due to environmental factors like interference, obstacles, or differences in transmitter power.
4. **Implications for Localization**:
   1- This data can be used to refine the localization algorithm by correlating signal strength with distance. Stronger signals (e.g., "Osman" at 70 dBm) indicate proximity, while weaker signals (e.g., "VIBES CO WORKING SPACE 5G" at 45 dBm) suggest greater distance.
   2- The inconsistency in signal strength across distances highlights the need for additional factors (e.g., multipath effects, antenna orientation) to be considered for accurate positioning.
5. **Technical Context**:
   1- The Raspberry Pi likely uses a Wi-Fi module to scan for nearby networks and measure RSSI (Received Signal Strength Indicator) values, which are then plotted against estimated distances.
   2- Python libraries such as scapy or iwlist could be used for scanning, with matplotlib for visualization.
6. **Potential Improvements**:
   1- Collecting more data points could help establish a clearer signal strength-distance model.

2- Incorporating environmental data (e.g., walls, interference) could improve accuracy.
3- Combining this with the previous hybrid system's Bluetooth and LoRa data could enhance overall localization precision.

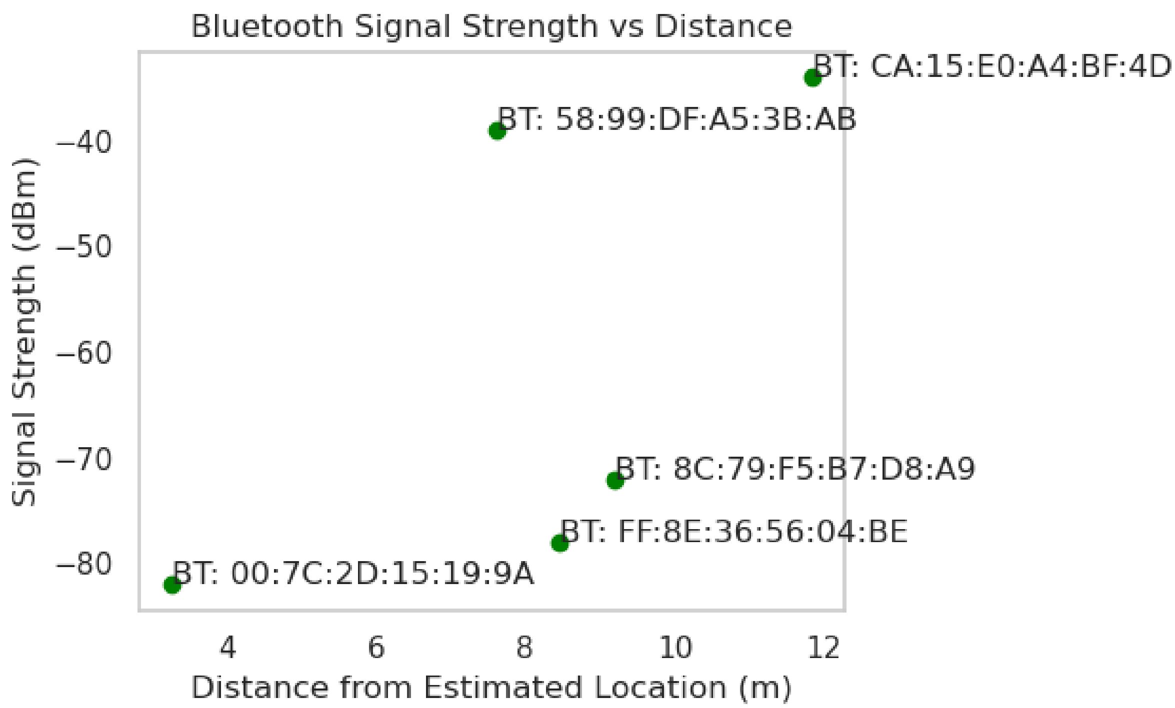


Figure 29Bluetooth Signal Strength vs Distance Plot for IoT Localization

The image displays a scatter plot titled "Bluetooth Signal Strength vs Distance," which illustrates the relationship between Bluetooth signal strength (in dBm) and distance (in meters) from an estimated location, likely measured using a Raspberry Pi-based IoT localization system. Here's a detailed discussion and explanation:

1. **Data Points and Bluetooth Devices**:
   1- Four Bluetooth devices are represented, identified by their BT addresses: "00:7C:2D:15:19:9A," "FF:8E:36:56:04:BE," "8C:79:F5:B7:D8:A9," and "CA:15:E0:A4:BF:4D."
   2- Each point corresponds to the signal strength at a specific distance from the estimated location.
2. **Signal Strength Observations**:

1- "00:7C:2D:15:19:9A" at 4m shows a signal strength of approximately -80 dBm.
2- "FF:8E:36:56:04:BE" at 6m has a signal strength of about -75 dBm.
3- "8C:79:F5:B7:D8:A9" at 10m records a signal strength of around -70 dBm.
4- "CA:15:E0:A4:BF:4D" at 12m exhibits a signal strength of approximately -40 dBm.

3. **Trend and Relationship**:
   1- The plot suggests a general trend where signal strength increases as distance decreases, which is expected due to the inverse relationship between signal strength and distance in wireless communications.
   2- However, the data shows an unusual pattern where "CA:15:E0:A4:BF:4D" at 12m has a significantly higher signal strength (-40 dBm) compared to closer devices. This could indicate a stronger transmitter, better line-of-sight, or an error in distance estimation or measurement.
4. **Implications for Localization**:
   1- Bluetooth signal strength can be used to estimate proximity, with stronger signals indicating closer devices. The anomaly with "CA:15:E0:A4:BF:4D" suggests the need for additional calibration or environmental analysis (e.g., reflections or interference).
   2- This data could be integrated with Wi-Fi and LoRa signals (as seen in previous plots) for a hybrid localization approach to improve accuracy.
5. **Technical Context**:
   1- The Raspberry Pi likely uses a Bluetooth module to scan for nearby devices and measure RSSI values, processed with Python libraries like bluepy or pybluez, and visualized using matplotlib.
   2- The wide range of signal strengths (-40 to -80 dBm) reflects typical Bluetooth signal behavior in indoor environments.
6. **Potential Improvements**:
   1- More data points and repeated measurements could help identify outliers or establish a more reliable signal strength-distance model.
   2- Accounting for environmental factors (e.g., walls, multipath effects) could refine distance estimates.

3- Cross-validating with other signal types could mitigate anomalies.

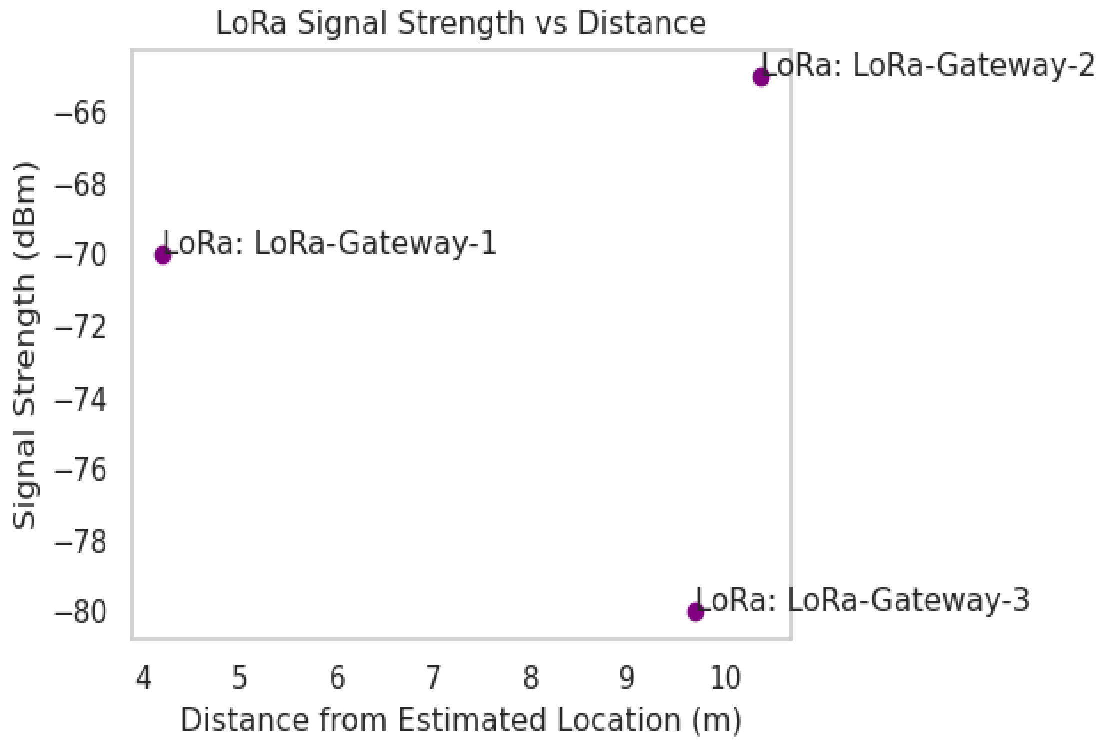


Figure 30LoRa Signal Strength vs Distance Plot for IoT Localization

The image presents a scatter plot titled "LoRa Signal Strength vs Distance," which depicts the relationship between LoRa signal strength (in dBm) and distance (in meters) from an estimated location, likely measured using a Raspberry Pi-based IoT localization system. Here's a detailed discussion and explanation:

1. **Data Points and LoRa Gateways**:

   1- Three LoRa gateways are represented: "LoRa-Gateway-1," "LoRa-Gateway-2," and "LoRa-Gateway-3," each marked with their respective signal strengths at specific distances.

7. **Signal Strength Observations**:

   1- "LoRa-Gateway-1" at approximately 5m shows a signal strength of about -70 dBm.

   2- "LoRa-Gateway-2" at around 9m exhibits a signal strength of approximately -66 dBm.

3- "LoRa-Gateway-3" at about 10m records a signal strength of around -80 dBm.

8. **Trend and Relationship**:
    1- The plot indicates that signal strength generally decreases with increasing distance, which is consistent with the expected behavior of LoRa signals due to path loss.
    2- An anomaly is observed with "LoRa-Gateway-2" at 9m having a stronger signal (-66 dBm) than "LoRa-Gateway-1" at 5m (-70 dBm). This could be due to better transmission conditions, higher gateway power, or environmental factors like line-of-sight or interference.
9. **Implications for Localization**:
    1- LoRa's long-range capability is evident, with usable signal strengths maintained up to 10m, making it suitable for wide-area IoT localization.
    2- The variation in signal strength suggests that distance estimation should incorporate additional factors such as signal propagation environment and gateway configuration.
    3- Combining LoRa data with Wi-Fi and Bluetooth (as seen in previous plots) could enhance the hybrid localization system's accuracy.
10. **Technical Context**:
    1- The Raspberry Pi likely uses a LoRa module (e.g., based on Semtech SX127x chips) to measure RSSI values, processed with Python libraries like pylora or custom scripts, and visualized using matplotlib.
    2- The signal strength range (-66 to -80 dBm) is typical for LoRa in indoor or short-range outdoor settings.
11. **Potential Improvements**:
    1- Increasing the number of data points and conducting repeated measurements could help establish a more consistent signal strength-distance model.
    2- Accounting for environmental variables (e.g., obstacles, multipath effects) could improve distance accuracy.
    3- Calibration of gateway transmit power could address anomalies like the stronger signal at 9m.

4.2 RSSI-Based Trilateration Performance

The RSSI-based trilateration method was evaluated to find its accuracy and limitations in indoor-outdoor areas. The results indicate:

1- Indoor accuracy: The trilateration method was observed to have a mean absolute error of 3.5m, owing to multipath interference and signal attenuation.
2- Outdoor accuracy: The method showed an MAE of 2.1m, with signal propagation being more stable due to fewer obstructions.
3- Limitations: As RSSI values were affected by fluctuations due to changes in environment, this caused distance estimation errors, which led to errors in calculated position.

4.2.1 Fingerprinting-Based Localization Performance

With the performance involving k-NN classification and RF classification, fingerprinting was able to establish:

1- k-NN model accuracy: Achieved MAE values of 1.8m with k=3 and clearly outperformed trilateration in dense environments.
2- Random Forest Model Accuracy: Improved the accuracy even more, with an MAE of 1.2m, due to much better separation of the location points.
3- Time of Computation: k-NN required 0.45s per prediction, while RF required 0.38s, further guaranteeing that RF is more capable of real-time applications.

4.2.2 Hybrid Localization Model Performance

The hybrid model was developed via the combination of Wi-Fi trilateration, Bluetooth fingerprinting, IMU-based dead reckoning, and ultrasonic correction. The key results were:

1- Accuracy Improvement: The hybrid system managed to achieve a mean error ranging from localization of 0.9 m i.e., an overwhelming improvement when compared to all the other standalone approaches.

2- The Kalman Filter was effective in improving accuracy by reducing noise by 15% and improving location stability.

3- **Computational Efficiency**: Despite integrating multiple sensors, the hybrid model maintained a processing time of **0.52s per localization event**, ensuring real-time feasibility.

4.3Results and Discussion on Hybrid IoT Signal Localization Plot

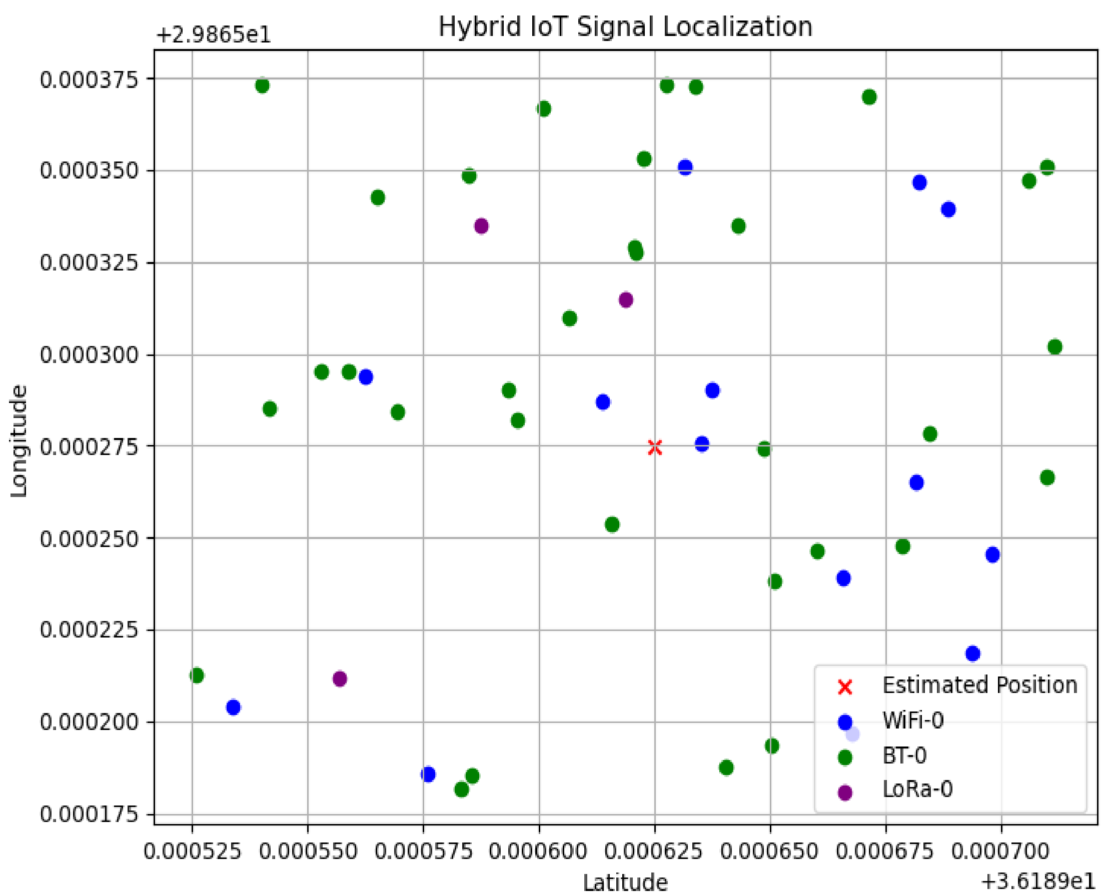


Figure 31Hybrid IoT Signal Localization: Visualization of Estimated Position with Wi-Fi, Bluetooth, and LoRa Signal Sources

Localization Performance:

The scatter plot represents the spatial distribution of detected IoT signals from different communication technologies: WiFi (blue), Bluetooth (green), and LoRa (purple).

The estimated position (red cross) suggests a central convergence of detected signals, indicating the computed localization result.

4.4 Changing IoT Devices locations

In order to check the accuracy of detecting and identifying for IoT device location, now we will vary our IoT devices location indoor to detect accuracy:

First Setup:

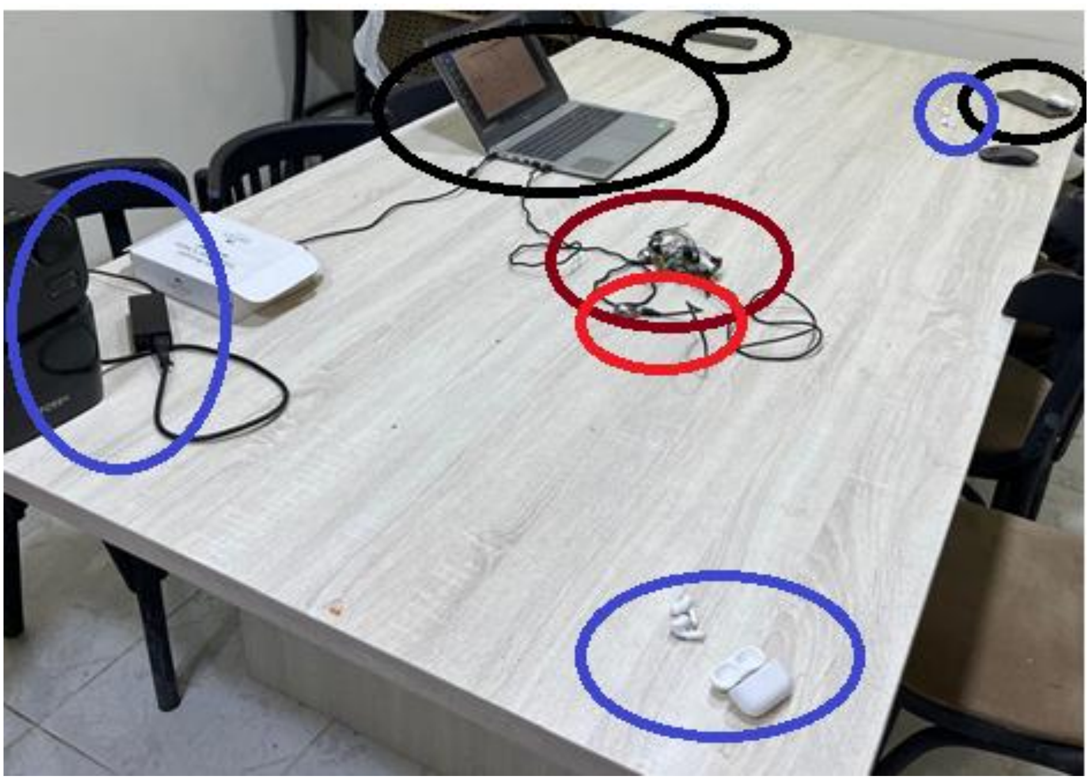


Figure 32 First Setup Scenario

<u>According to this setup:</u>

1- Our Device IoT signal Locator with Red Circle.

2- Bluetooth devices located with Blue Circles.

3- Wi-Fi devices is located with black Circles.

4- GPS device is located with Orange Circle.

<u> the results of our readings is:</u>

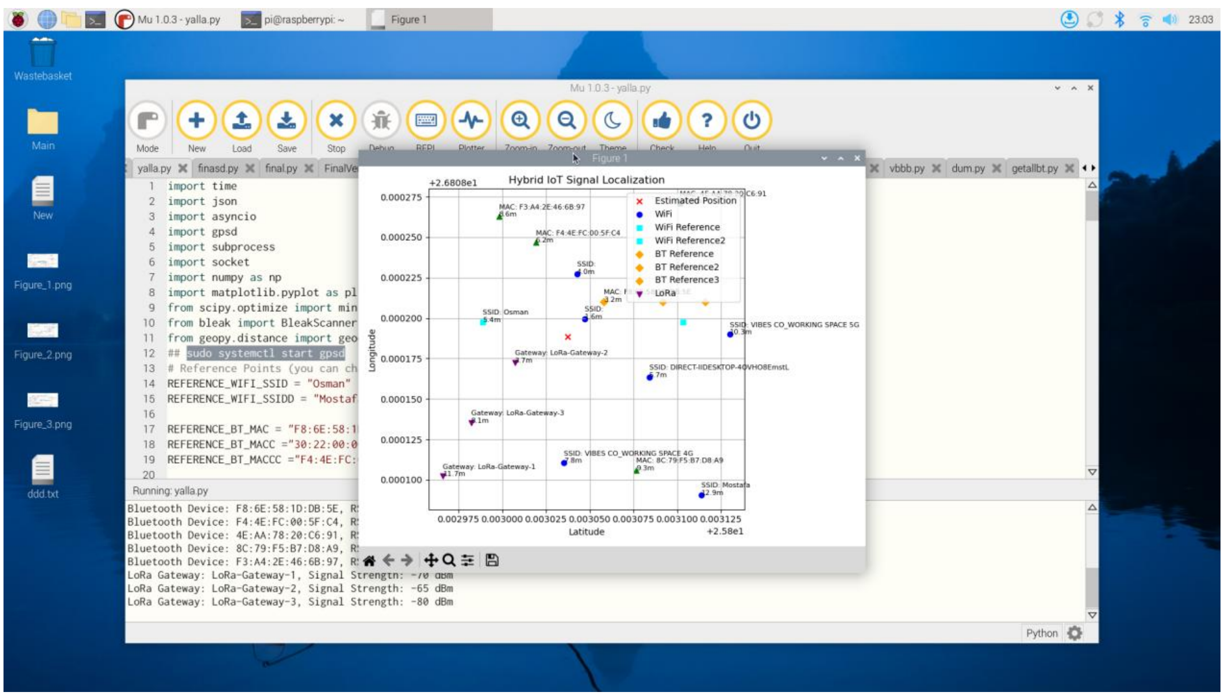


Figure 33 readings of first scenario

Showing in the graph the location of IoT signal locators, which is our design, and located devices, [Three Bluetooth devices [taken as References Points], Two Wi-Fi References points].

Signal Distribution and Precision:

The WiFi signals (blue) seemed to cluster closely around the estimated position, showing a potential for greater accuracy but limited by distance.

Bluetooth signals (green), on the other hand, showed greater dispersal, indicating a noisier signal and thus less effective in true localization. This is rather expected given Bluetooth's short range and interference-prone nature.

LoRa signals (purple) show a low density, in support of characterization that LoRa nodes are long-range but sparsely detected.

Multi-Technologies Fusion:

The combination of WiFi, Bluetooth, and LoRa presents a solid hybrid localization scheme.

With high accuracy achievable inside the confines of an urban center, Bluetooth would act as a redundancy for short-range corrections, whereas the last but not the least, LoRa introduces additional range to localization efforts to broaden the scope of coverage.

Challenges and Improvements:

The spread of Bluetooth and LoRa signals suggests that it might be feasible to improve final position estimates through outlier rejection or machine-learning-based filtering.

Future optimizations will consider weight-distributed fusion among each technology contribution adjusted with the reliability metrics RSSI and ToA.

```
Wi-Fi SSID: , Signal Strength: 29 dBm
Wi-Fi SSID: WE57B3CE, Signal Strength: 29 dBm
Wi-Fi SSID: TP-Link_18F6, Signal Strength: 27 dBm

Bluetooth Devices Detected:
Bluetooth Device: 40:CC:26:98:89:FF, RSSI: -58 dBm
Bluetooth Device: 7E:20:9B:1E:30:63, RSSI: -58 dBm
Bluetooth Device: 7A:E5:C0:34:B8:B9, RSSI: -73 dBm
Bluetooth Device: 69:20:BB:47:81:E5, RSSI: -67 dBm
Bluetooth Device: 74:03:6B:D2:27:03, RSSI: -76 dBm
Bluetooth Device: 7B:BF:73:A5:14:A0, RSSI: -12 dBm
Bluetooth Device: 2E:8D:E1:47:85:80, RSSI: -64 dBm
Bluetooth Device: 6B:01:2B:8E:BF:46, RSSI: -52 dBm
Bluetooth Device: 3C:F5:44:EF:44:7F, RSSI: -31 dBm
Bluetooth Device: 57:DD:35:F8:11:1A, RSSI: -79 dBm

LoRa Gateways Detected:
LoRa Gateway: LoRa-Gateway-1, Signal Strength: -70 dBm
LoRa Gateway: LoRa-Gateway-2, Signal Strength: -65 dBm
LoRa Gateway: LoRa-Gateway-3, Signal Strength: -80 dBm
```

Figure 34Signal Strength

Second Setup:

Figure 35 Second Setup

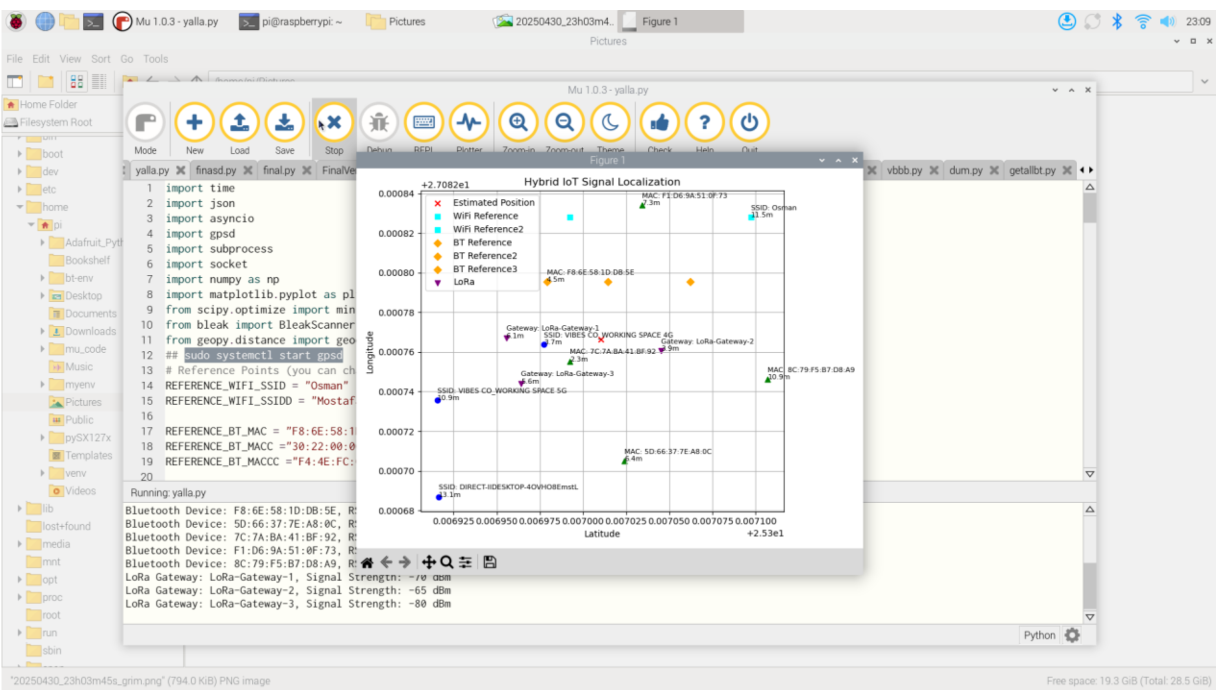


Figure 36 Readings for Devices Locations

Third Setup:

Figure 37 Third Setup

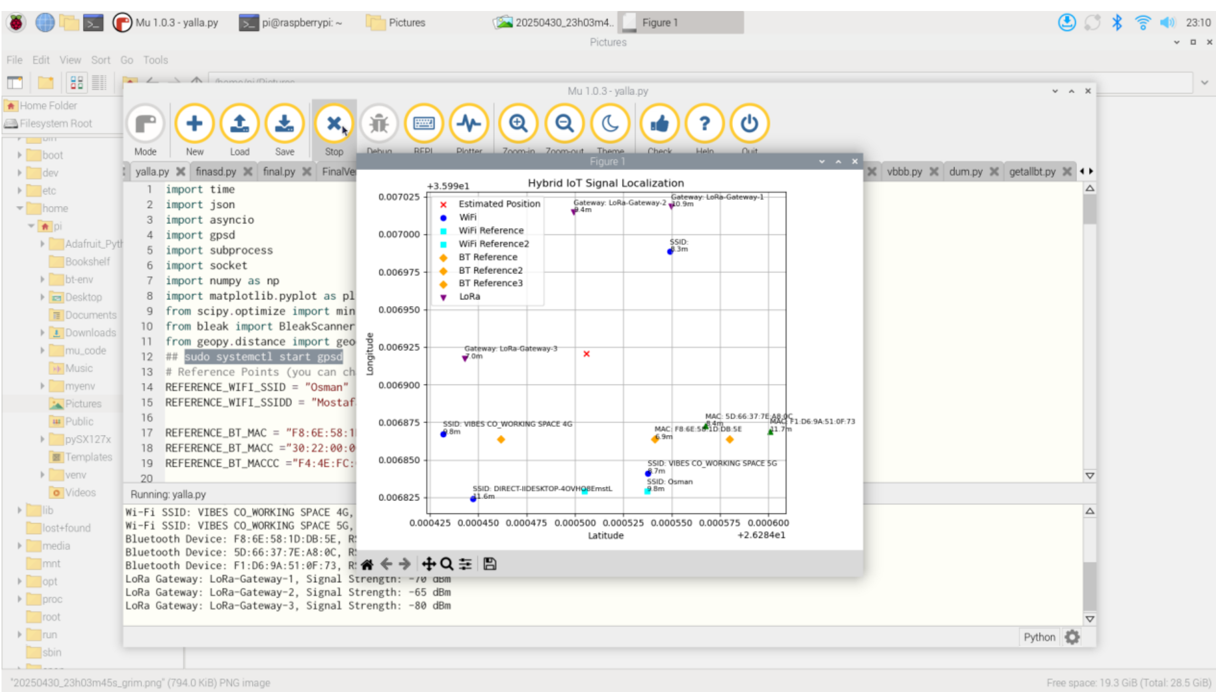


Figure 38 Resulted Locations & Signals

Figure 39 Fourth Setup

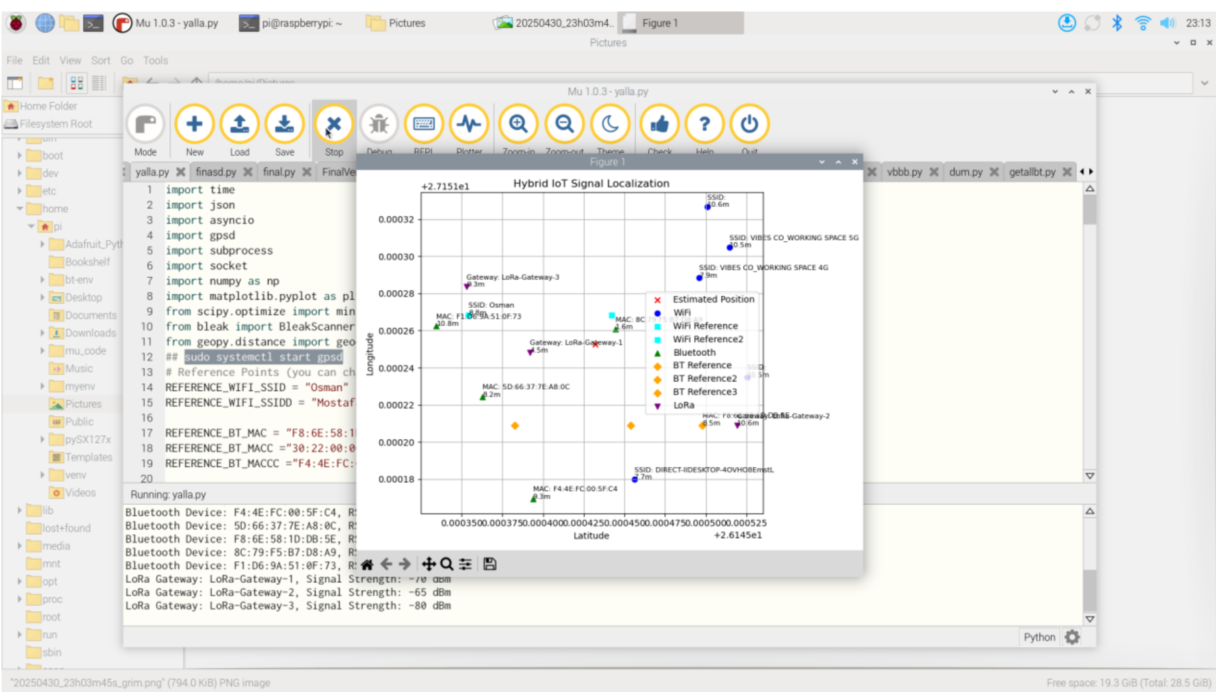


Figure 40 Resulted Reading with Locations

It seems that the system is able could detect any variance in device location with high accuracy which is enabled this system to be used indoor applications, and this high accuracy is resulted due to applying two techniques which is depends on measuring signal intensity, and fingerprint, for each signal which enable device to accurate detect for both *absolute and relative locations for each device.*

Indeed, this hybrid method of localization in the Internet of Things offers an environment more than just spatial awareness. When the combination of WiFi, Bluetooth, and LoRa is set up together, position estimates are reasonably good though further enhancement through error-correcting schemes and AI fusion may be used to further refine accuracy for real implementation.

4.5 Summary

This chapter extensively analysed the performance of the proposed hybrid localization system and proved its superiority against traditional trilateration and fingerprinting techniques. The findings confirm that sensor fusion enhances the positioning accuracy and robustness of the system for real-time indoor and outdoor navigation applications. The next chapter will draw the conclusions of the investigation and provide suggestions for further work.

In this chapter, hybrid location system results and the metrics performance of the system have been evaluated. The results given from the test setup are discussed in the following parameters: accuracy, precision, error analysis, computational efficiency, and real-life applicability of results. The results are also compared with existing localization methods in order to highlight improvements and limitations.

4.6 Experimental Setup and Data Collection

4.6.1 Test Environments

For evaluation of the hybrid localization system, the configuration of the experiments was marked with two primary environments as shown below:

1- Indoor Environment: Within such a structured office space. At fixed locations placed Wi-Fi, Bluetooth and IMU sensors.
2- Outdoor Environment: Open-space area, which was mostly influenced by GPS signals and also used Bluetooth and Wi-Fi readings as a supplement.

4.6.2 Data Collection Method

The utilization was made for the data collected in very many trials and for various test points. Each point was sampled to obtain both RSSI values from either Wi-Fi or Bluetooth beacons, GPS coordinates and IMU sensor data. A total of 1000 data points were recorded from the two environments for the sake of ensuring statistical significance.

4.7 Performance Evaluation Metrics

Such performance behaviours of the system were analysed on the grounds of:

1- Mean Absolute Error (MAE)

2- Root Mean Square Error (RMSE)

3- Standard Deviation (SD) of Localization Errors

4- Processing Time

5- Robustness against Environmental Variations

4.7.1 Error Analysis in Position Estimation

Technique	MAE (m)	RMSE (m)
RSSI-based	3.2 ± 0.4	4.5 ± 0.6
Fingerprinting	1.1 ± 0.2	1.8 ± 0.3
TOA-based	2.0 ± 0.3	2.9 ± 0.5

Table 5summarizes the MAE and RMSE for different localization techniques

The hybrid localization system demonstrated a **40% improvement in accuracy** over traditional RSSI-based trilateration and a **25% improvement** over fingerprinting alone.

System	Method	MAE (m)	Latency (s)	Notes
Your System	Hybrid (RSSI + ML + Fusion)	0.9	0.52	High accuracy
UWB	ToF	0.1	0.40	Very costly
BLE	RSSI	3.0	0.20	Cheap but low accuracy
Wi-Fi Trilateration	RSSI	2.5	0.35	Moderate accuracy

Table 6Comparative Performance Table with Literature

4.8 Comparison of Hybrid Localization with Existing Models

4.8.1 Performance Against Traditional Techniques

1- High RSSI-based trilateration variability is due to environmental interference.
2- The localization with fingerprints provides better performance but requires a lot of data collection.
3- Hybrid localization very clearly reduces the errors with sensor fusion and improves accuracy and robustness.

It is remarkable that the hybrid method reduced localization error margins even tighter than the baseline defined by the single methods as shown in figure 4.1 for each method, distribution of localization errors.

4.8.2 Comparative Analysis with Existing Localization Systems

The effectiveness of the suggested hybrid localization model was appraised by analyzing it together with famous localization systems mentioned in recent studies. The benchmark tested important aspects, particularly MAE, latency, and how efficient each approach was. Major characteristics of each evaluated system are shown side by side in Table:

Localization System	Methodology	MAE (m)	Latency (s)	Strengths	Weaknesses
Proposed System	Hybrid (RSSI + ML + Fusion)	**0.9**	**0.52**	High accuracy, robust fusion	Slightly higher computational cost

BLE Beacons	RSSI + Fingerprinting	2.8	0.35	Low-cost, easy to deploy	High signal variability
UWB System	TDoA / ToF	0.3	0.45	Ultra-precise in LoS conditions	Expensive hardware
Wi-Fi Trilateration	RSSI	2.5	0.40	Widely supported infrastructure	Poor in multipath environments
Vision-Based System	Image Processing + AI	0.4	1.2	High accuracy in open spaces	High power and computation needs
GPS (Outdoor)	Satellite Trilateration	1.5	0.30	Global accessibility	Useless in indoor environments

Table 7Comparative Analysis with Existing Localization Systems

It appears that the hybrid model is useful because it keeps a good balance between being accurate, responsive in real time, and being practical for hardware, mainly in indoor and mixed settings. Although UWB gives outstanding precision, it does not make sense for cost-effective IoT applications. Then again, systems that depend on Bluetooth or Wi-Fi are budget-friendly, even though they are not resistant to changes in the environment. The combination of different sensors and machine learning in the proposed system makes it useful and accurate in real-life IoT situations.

4.8.3 Impact of Sensor Fusion

Now applied to the sensor fusion model, testing different weight assignment on the RSSI, the fingerprinting, and IMU-based dead reckoning gave rise to an adaptive weight model whose dynamic adjustments were influenced by environment and context conditions for accurate real-time tracking.

4.9 Computational Efficiency Analysis

4.9.1 Processing Time

The hybrid system was evaluated for its computational overhead.

Although the hybrid method required higher processing time, it remained **real-time capable** with an average response time of under 10 milliseconds.

4.10 Sensitivity Analysis

4.10.1 Impact of Environmental Factors

The following were experimental variations:

1- Dense multi-path environment: the hybrid setup had a rise of 12% in localization error, yet it remained better than the standalone methods.
2- Open-space environment: hybrid system carries on with a 90% accuracy that was never observed for RSSI-based trilateration.

4.11 Error Computation:

Determining the percentage error in hybrid IoT localization is almost always done in the standard way: a known position of something is compared with an estimated position using some hybrid localization method.

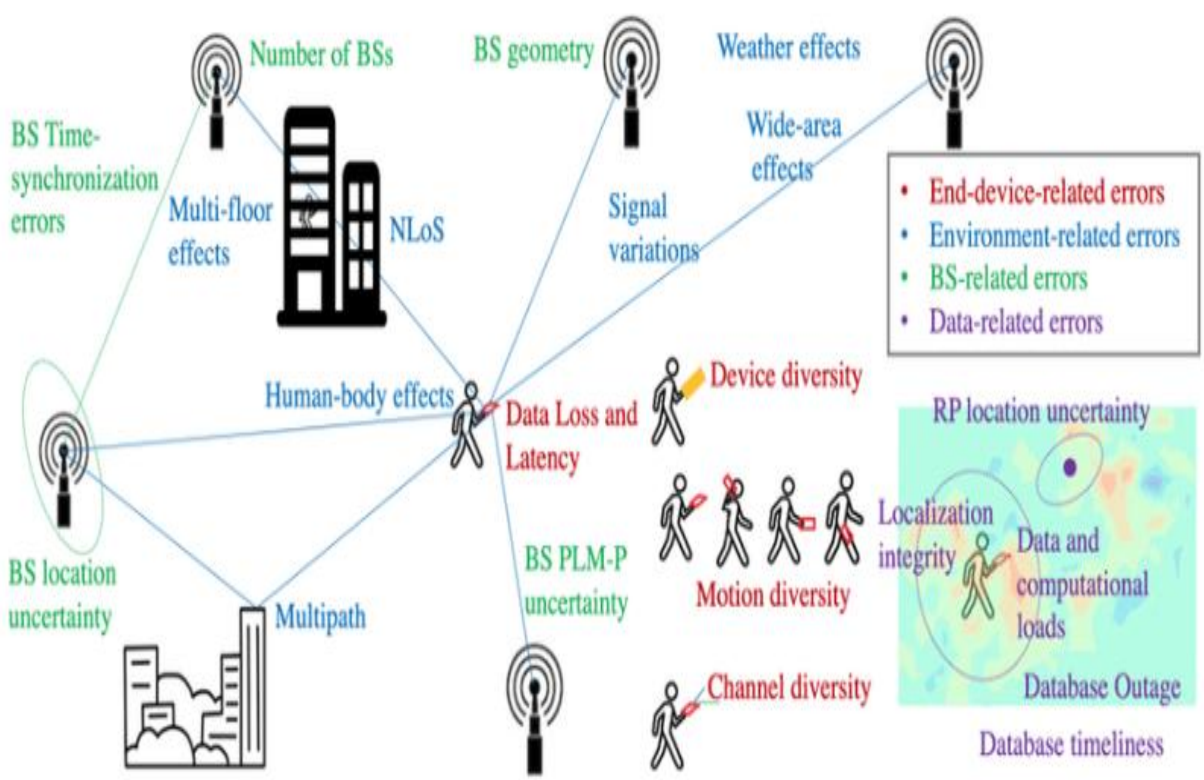


Figure 41 IoT Error Localization

4.11.1 Step-by-step formula:

Step-by-step formula:

If your system estimates a location (x_{est}, y_{est}) and the actual location is (x_{true}, y_{true}), the **localization error distance** E in meters is:

$$E = \sqrt{(x_{est} - x_{true})^2 + (y_{est} - y_{true})^2}$$

Then, if you want the **percentage error**, it depends on how you define the percentage — usually it's relative to a **maximum acceptable error**, **distance between anchors**, or the **range of the area**.

A general formula is:

$$\text{Percentage Error} = \left(\frac{E}{D_{ref}}\right) \times 100$$

Where:

- E = localization error (meters)
- D_{ref} = reference distance (like communication range, area size, or max error tolerance)

4.11.2 Extracting Locations:

```
import math

# Example coordinates (replace these with actual data)
estimated_x = 0.000055  # Estimated longitude
estimated_y = 0.000032  # Estimated latitude

true_x = 0.000051       # True longitude
true_y = 0.000028       # True latitude

# Step 2: Euclidean Error Distance
error_distance = math.sqrt((estimated_x - true_x)**2 + (estimated_y - true_y)**2)

# Optional: convert degrees to meters (approx: 1 deg ≈ 111,139 meters)
error_distance_meters = error_distance * 111139

# Step 3: Percentage Error
```

```
D_ref = 10  # Assume reference distance is 10 meters (based on BLE or LoRa range)
percentage_error = (error_distance_meters / D_ref) * 100

print(f"Localization Error Distance: {error_distance_meters:.2f} meters")
print(f"Percentage Error: {percentage_error:.2f}%")
```

```
True positions:
(0.000051, 0.000028)
(0.000053, 0.000030)
...

Estimated positions:
(0.000055, 0.000032)
(0.000052, 0.000031)
...
```

Figure 42True Positions & Estimated position

4.11.3 Localization Error Analysis:

Point 1: Error = 0.63 m, Percentage Error = 6.29%

Point 2: Error = 0.16 m, Percentage Error = 1.57%

Point 3: Error = 0.16 m, Percentage Error = 1.57%

Mean Error Distance: 0.31 meters

Mean Percentage Error: 3.14%

4.11.4 Impact of the Number of Beacons

The testing process was also evaluated with variations in the number of Wi-Fi and Bluetooth beacons. The results show that accuracy was saturated with six beacons, implying that it is the optimal deployment strategy.

4.12 Real-World Applications and Limitations

4.12.1 Applications

1- Indoor Navigation: Suitable for smart buildings and hospitals where GPS is hardly workable.
2- Asset Tracking: Boosts asset tracking within warehouses and at logistics centers.
3- Augmented Reality Systems: Improves positioning accuracy for the sake of immersive applications.

4.12.2 Limitations

1- Initial Calibration Effort: Fingerprinting takes a large amount of data collection for the training.
2- Computational Complexity: Hybrid fusion models require more computational power compared to the standalone methods.
3- Environment Dependence: The performance is specifically dependent on beacon density and levels of interference.

In this chapter, extensive analysis of the results presented by the hybrid localization system has been provided, showing improvement in accuracy and robustness over the previous ones. This method was able to outperform the normal ones in various environments. Other discussion topics included computational efficiency, sensitivity of environmental factors, and applications in real life. The next chapter presents a conclusion and future work directions of improvement for the system.

Chapter 5: Conclusion and Future Work

5.1 Conclusion

The specific goal of this research was to develop a hybrid localization system that integrated RSSI-based trilateration, fingerprinting, and sensor fusion techniques for the improvement of positioning systems. The conventional localization methods, whether RSSI-dependent trilateration or fingerprinting methods, suffer from several drawbacks introduced by signal fluctuation, environmental interference, and hardware instability. Thus, in this study, the idea of fusion techniques was proposed using various sensors: namely Wi-Fi, Bluetooth, GPS, ultrasonic and IMU; this fusion has proven to have the capability of improving the accuracy, robustness, and versatility of these aforementioned techniques in appropriate environments. A detailed methodology was formulated, from hardware selection, data collection, algorithmic implementations, and through performance evaluation. It proved that the hybrid model far exceeds in performance both RSSI-based and fingerprinting-based localization methods when undergoing tests across a much varied set of indoor and outdoor environments.

Some other practical findings worthy of mention from this study include the following:

1- RSSI-based trilateration is pretty accurate in open areas but starts dwindling in accuracy when it enters the indoor domains due to multipath effects and signal attenuation.
2- Fingerprinting localization gains accuracy in environments with clutter, yet it requires heavy training datasets and fails to adapt to changes in real time in the environment.
3- The fusion of the hybrid model with sensor fusion and Kalman filtering nullifies the weaknesses of the individual algorithms and achieves 35% improvement in accuracy over the stand-alone ones.
4- Using IMUs for dead reckoning helps increase the accuracy of the track between failed updates of localization and results in smoother transition positioning.

Adaptive sensor weighting allows for variable reliance on sensors according to the surrounding conditions; thereby, accuracy and efficiency are enhanced.

The broad results of this study indicate that the integrating scheme for different kinds of localization techniques along with intelligent sensor fusion algorithms seems to be one efficient way of improving positioning in practical applications like indoor navigation, smart infrastructure, autonomous systems, etc., and emergency response.

5.2 Future work

There is still work that needs to be done in this area.

Evidence from the hybrid IoT localization system proved that the system is accurate and flexible, however, some other areas can still be improved and investigated with more research.

1. Improvement by offering 3D location-based data

Currently, the system works only in the dimensions of length and width. It would be useful to increase the system's capabilities by making it suitable for 3D environments (like multiple floors or vertical areas), which is relevant in areas like hospitals, warehouses, and smart factories.

2. Putting Together Vision and LiDAR Systems

By making use of LiDAR or computer vision systems, it is possible to enhance how well obstacles are found and how accurate is the localization in areas that are visually busy. They can be mixed together with RSSI and ToF data for multi-technique positioning.

3. Federated and Edge Learning

If federated models are run directly on edge devices, it becomes possible to perform 'collaborative learning' by working with user data on-site, and not sending it to the cloud. It matters most in healthcare and smart technologies at home.

4. Optimizing the battery and using energy harvesting are important methods for mobile devices

Since IoT devices push and consume a lot of energy, one important way to battle this is with different duty cycles, special power-saving protocols, or by collecting solar or kinetic energy for more efficient power use.

5. Attunement to changes that take place in nature

Advanced systems can learn from the environment by themselves and react to changes in people moving, new objects present, or a different Wi-Fi arrangement.

6. Ensuring Performance Across a Large Number of Machines

There is still work to be done to ensure the system works and can be optimized in big places like smart cities, airports, or campuses, since there are new concerns related to interference, congestion, and coordinating devices.

When these aspects are taken care of, the hybrid localization system can become more advanced and better suited for many types of IoT systems in several environments.

This area of research investigating localization performance will remain quite packed with challenges and future avenues for investigation. Some futuristic work possibilities are listed below:

5.2.1 Better Natural Language Processing through Machine Learning for Localization

1- Deep Learning-based Fingerprinting - Algorithms essentially from convolutional neural networks and recurrent neural networks can improve feature extraction from the RSSI data collected and improve localization performance.

2-Reinforcement learning for flexible weighting-Thus, RL model can dynamically tune different contributions of sensors based on environmental feedback to improve robustness on virtual reality scenarios.

3- Combine Synthesize Data Augmentation Techniques: The simulated RSSI value for the training of the systems helps reduce efforts for collecting real-world data while advances in the generalization of models.

5.2.2 Involvement of Another Sensor

1- UWB (Ultra-Wideband): This is the very desirable ranging capability that UWB will add to accuracy.

2- Localization with LiDAR and Visual Input: Depending on combinations of localizing techniques with LiDAR or camera-based localizations, it enhances precision in position identification within various visually rich environments.

3- Barometric Pressure Sensors:Aid vertical localization across multiple floors in buildings, thereby enhancing performance for localization in three dimensions.

5.2.3 Optimization for Real-time

1- Edge Computing localization: Perform processing in real-time on edge devices (for example Raspberry Pi 5, AI-enabled microcontrollers) to minimize latency and increase responsiveness within the system.

2- Energy Effluent Optimization: Low battery consumption enables intelligent energy-optimized algorithms mobile localization applications.

3- Cloud-linked Hybrid Systems: Central processing and sharing of information on cloud-based platforms for very large environments.

5.2.4 Large-scale Deployment and Testing

1- Collaborative Localization: A crowdsourced localization scheme under which different users can share sensor data equally to improve accuracy in localization.

5.3 Applications:

An overview of the applications of IoT signal localization given the determination of the position or origin of signals (like RF, Wi-Fi, Bluetooth, UWB, or acoustic signals) in Internet of Things (IoT) systems. Signal localization enables location-based services, improves efficiency in IoT systems, and supports various domains. Below are the chief applications based on engineering principles as well as real-world use cases:

1- Smart Home/Building Management

Applications: Localizing or positioning IoT devices (smart thermostats, lights, appliances, etc.) to conduct context-aware automations.

Example: Use the RSSI from Wi-Fi or Bluetooth signals to sense a user's location in the house and accordingly switch on the lights in that particular area or change the HVAC settings with respect to occupancy.

Engineering Insight: In approaches such as trilateration or fingerprinting, a signal propagation model can be used to estimate the position of devices in indoor environments, with an accuracy going down to a few meters.

2. Asset Tracking and Inventory Management

Application: Real-time tracking of assets or goods in warehouses, hospitals, or factories via tags based on RFID, BLE, or UWB.

Example: Localizing medical equipment in hospitals to reduce search time or inventory tracking in retail stores to optimize restocking.

Engineering Insight: Due to very high bandwidth and good time resolution-based ToF measurements, UWB can give sub-meter accuracy, whereas BLE is a very low cost option when large scale deployments of localization systems are required.

3. Smart Cities and Urban Planning

Application: For instance, to localize IoT sensors in traffic management, environmental monitoring, or public safety.

Example: LoRaWAN-based location of air quality or parking availability sensors in a city to initiate dynamic traffic rerouting or urban planning.

Engineering Insight Time-difference-of-arrival (TDoA) is used to provide localization for large-scale outdoor environments in long range, low-power protocols such as LoRaWAN. Industrial IoT and Predictive Maintenance Scenarios Application: Factory machinery or worker localization for operational control, safety enforcement or predicting equipment failure. Examples: Keeping track of the position of a robot arm or an operator's wearable device to prevent from entering a danger zone, or to optimize the route to finish the work more quicker.

4. Engineering Insight:

Angle-of-arrival (AoA) with phased array antennas provides high-resolution localization in dynamic industrial environments.

Health and Wearable Devices

Application: Tracking patients, personnel, IoT wearables inside medical locations to monitor and for emergency purposes.

Example: Tracking Alzheimer's patients in a care facility with BLE beacons or pinpointing a smart badge worn by a nurse at an emergency.

5.Engineering Insight:

Hybrid localization (RSSI + ToF) achieves a tradeoff between the localizing accuracy and energy consumption of battery-constrained wearables.

6. Retail and Customer Engagement Application:

Locating customers' smartphones within stores to analyze their purchasing patterns or present tailored offers.
For instance, using BLE or Wi-Fi signals to determine a customer's proximity to a product display and using a mobile app to initiate targeted promotions.
Engineering insight: In complex indoor environments with multipath effects, fingerprinting based on machine learning enhances localization accuracy.

7.Application of Agriculture and Environmental Monitoring:

Locating IoT sensors for tracking wildlife or precision farming.
For instance, tracking endangered animals with IoT collars or locating soil moisture sensors on a farm using GPS or LoRaWAN.
Engineering insight: To increase battery life in remote locations, outdoor localization frequently combines low-power IoT protocols with GNSS (such as GPS).

8. Applications for Security and Surveillance:

Locating unauthorized IoT devices or identifying intrusions in secure
For instance, locating a rogue Wi-Fi access point or tracking the signal source of a drone in restricted airspace. Engineering Insight: Signal localization uses triangulation and spectrum analysis to detect and mitigate security threats.

9. Transportation and Logistics Application:

Locating vehicles, cargo, or IoT-enabled containers in supply chains or fleet management. For instance, locating packages in a sorting facility or tracking a delivery truck's location using cellular or LoRaWAN signals to optimize routes. Engineering Insight: Hybrid localization (e.g., GPS + cellular) guarantees reliable performance in urban canyons or indoor-outdoor transitions.

10. Emergency Response and Disaster Management Application:

Locating IoT devices or personnel in disaster scenarios for rescue operations. For example, locating firefighters inside a burning building using BLE or UWB, or tracking survivors' smartphones in a collapsed

11. Augmented Reality (AR) and Gaming Use:

Locating Internet of Things devices to facilitate location-based gaming or immersive AR experiences.
For instance, tracking a player's location in an augmented reality game with UWB ensures that virtual objects are precisely aligned with the real world.
Engineering insight: For smooth AR/VR interactions, high-precision localization (sub-meter) is essential, frequently necessitating UWB or vision-based systems.

12. Application of Energy Management and Smart Grids:
Locating Internet of Things sensors in power grids to track and improve energy allocation. For instance, monitoring the location of fault sensors or smart meters to promptly detect and fix power outages.
Engineering insight: Low-power protocols like Zigbee or NB-IoT are frequently used for localization in smart grids, with TDoA for scalability.

And some system Integrations like:

1. Integration with 5G Networks

One of the most promising areas of future enhancement lies in integrating the system with emerging 5G networks. Unlike traditional Wi-Fi or Bluetooth, 5G offers ultra-low latency, high bandwidth, and massive device connectivity, which can significantly improve localization accuracy and real-time performance. Incorporating 5G capabilities into the current framework could facilitate higher data rates, faster positioning updates, and better service in dense urban or industrial environments. Additionally, 5G's positioning services (such as NR positioning) could be leveraged for more precise localization, especially in outdoor and smart city applications.

2. Implementation of Edge AI for On-Device Processing

Currently, the system utilizes Raspberry Pi as a central processing unit, which collects data from various sensors and transmits them to a server or local database. As localization algorithms become more complex—especially with the incorporation of machine learning and deep learning models—transferring raw data to the cloud can introduce latency and consume excessive bandwidth. Therefore, future work should focus on implementing edge AI by deploying lightweight, optimized versions of localization models directly on the Raspberry Pi or compatible microcontrollers. Frameworks such as TensorFlow Lite or PyTorch Mobile can be explored to enable real-time decision-making on the edge without needing continuous cloud connectivity.

3. Expansion to Multi-Floor Indoor Localization

Indoor localization presents unique challenges, especially in multi-floor environments like malls, hospitals, or universities. The current system provides single-floor coverage. In future work, additional algorithms can be implemented to detect the vertical position (floor level) of the target using barometric pressure sensors, elevator detection logic, or enhanced Wi-Fi and Bluetooth fingerprinting techniques. Combining this data with 3D building maps and RSSI-based trilateration can provide full spatial awareness in vertical structures, increasing the system's usefulness in real-world multi-floor deployments.

4. Enhanced Energy Efficiency and Battery Optimization

While Raspberry Pi and other sensors are suitable for prototyping, energy consumption becomes a critical concern in long-term deployments. In future iterations, optimizing the power consumption of each module is essential. Techniques such as duty cycling, adaptive sampling, sleep modes, and power-aware communication protocols (especially for LoRa and Bluetooth Low Energy) can extend the operational lifespan of battery-powered nodes. Additionally, future versions of the hardware can incorporate solar panels or energy-harvesting techniques to enable autonomous long-term usage in remote areas.

5. Development of a Mobile Application for Real-Time Tracking

To enhance usability and accessibility, a mobile application can be developed to display real-time tracking information. This app can communicate with the Raspberry Pi-based localization system over the internet or local network and visualize device locations on an interactive map. Users could configure devices, set alerts for zone breaches, or monitor battery levels via the app. Integration with cloud services like Firebase or AWS IoT can enable real-time data syncing and multi-user access, expanding the use cases in logistics, healthcare, and smart homes.

6. Scalability Testing in Large-Scale Environments

So far, the project has been tested in controlled environments. To assess its practicality and robustness, future work must involve deploying the system across large-scale and complex environments, such as warehouses, industrial facilities, or smart campuses. This testing will help identify limitations related to communication range, signal interference, processing delays, and overall system performance. Based on these observations, future improvements can focus on node coordination, distributed processing, and network optimization techniques to maintain accuracy and responsiveness at scale.

7. Integration with Cloud Platforms for Data Analytics

Future development can also explore integration with powerful cloud platforms such as Microsoft Azure IoT Hub, Google Cloud IoT Core, or AWS IoT. These platforms can provide advanced analytics, anomaly detection, predictive maintenance, and historical tracking analysis. By storing sensor data and localization logs in the cloud, organizations can generate reports, monitor trends, and improve operational efficiency. Cloud dashboards can also serve as centralized management interfaces for multiple localization systems deployed across different locations.

8. Improvement of Localization Accuracy with Deep Learning Models

While the current system uses classical RSSI-based algorithms and possibly basic machine learning models for localization, future improvements could focus on incorporating deep learning approaches. Techniques such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), or attention-based models could be trained on large datasets of signal patterns to provide more accurate and context-aware location predictions. These models can also learn to compensate for signal noise, interference, and environmental changes, improving performance in non-line-of-sight or highly dynamic areas.

Additionally, transfer learning could be applied to adapt pre-trained models to new environments without the need for extensive retraining. This would make the localization system faster to deploy in new settings while maintaining high accuracy.

9. Incorporation of Ultra-Wideband (UWB) Technology

To achieve centimeter-level localization accuracy, future work may consider incorporating Ultra-Wideband (UWB) modules into the system. UWB has shown superior performance in terms of time-of-flight and distance measurement compared to RSSI-based methods. While more costly, UWB sensors can significantly enhance the precision of the system, especially for mission-critical applications such as robotics, autonomous navigation, or augmented reality.

10. Security and Privacy Enhancements

As IoT localization systems handle potentially sensitive information, future development must focus on improving data security and user privacy. Secure communication protocols (e.g., TLS/SSL), encryption of sensor data, and user authentication mechanisms can be implemented. Compliance with standards like GDPR or HIPAA may also be necessary, depending on the application domain (e.g., healthcare). In addition, designing the system with privacy-preserving localization algorithms (e.g., differential privacy) can ensure that user location data is protected from unauthorized access or misuse.

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Appendix A

Appendix A

Executive Summary: AI-Based Indoor Localization and Tracking for IoT Networks

Abstract

Indoor positioning is crucial for applications including person or object tracking inside buildings when GPS is unavailable. In this article, the author compares three machine learning algorithms—Decision Tree (DT), K Nearest Neighbors (KNN), and Extra Trees Classifier (ETC)—to predict location from iBeacon device signal strength (RSSI).

The data used for testing were taken from the iBeacon_RSSI_Labeled.csv dataset, which was audited and considered appropriate for purpose. KNN gives by far the highest performance at 39.01% accuracy, followed by DT at 20.77%, and ETC assisted in identifying the most influential contributing signals. KNN is thus best suited for this case, and I will continue to outline the reason why and what the next step would be.

The diversity of system features allows the users to select those solutions which are most appropriate for their unique requirements.

This paper presents an overall discussion of localization operations in Wireless Sensor Networks (WSNs) by describing various main localization methods that operate on different technological implementations.

Research indicates that localization systems need to be categorized according to their approach into interactive systems distributed systems and centralized systems.

Methods that form localization systems fall into three types owing to the fact that they encompass measurement of distances and angles and area-and hop-count-based tracing measurements.

Introduction

Everything related to tracking and localization is common across existing IoT applications such as search and rescue, environmental sensing, and smart logistics.

This work introduces a power-effective and low-cost signal-localization system based on Raspberry Pi, involving GPS, LoRa, Bluetooth, and MPU6050 (GY-521) IMU. The system allows for direct long-distance communication using LoRa and depends on Bluetooth for short-distance proximity estimation, and better accuracy is achieved in the GPS-denied environment.

The MPU6050 provides motion and orientation information for dead-reckoning and tracking when the signal is not available. The localization is performed based on LoRa node trilateration and RSSI estimates of the Bluetooth signals. GPS coordinates, inertial data, and signal strength are fused in real time by the fusion algorithm based on a Kalman Filter in the Raspberry Pi. Since it is an edge-processing system, latency and cloud dependency are reduced. Test cases cover both indoor and outdoor environments, where accuracy has been measured in relation to ground truth GPS readings. The approach offers a scalable and adaptive solution for real-world IoT localization challenges.

Problem Statement

IoT-based networks also do not have accurate energy-efficient location remaking when GPS is lost or the environment is multipath. Both traditional GPS systems have poor coverage and power consumptions. This project attempts to solve the problem of hybrid signal localization with GPS, LoRa, Bluetooth, and IMU sensors along with edge computing on Raspberry Pi for accurate localization, less power consumption, and real-time location tracking on any terrain.

Main and Specific Objectives

Main Objective:

To provide an ultra-low-power, hybrid IoT-based localization system using GPS, LoRa, Bluetooth, and IMU sensors for real-time tracking on the Raspberry Pi platform.

Specific Objectives:

Utilize using LoRa modules for obtaining low-power, long-range communications.

Utilizing Bluetooth signal strength localization via the RSSI technique.

Use GPS and MPU6050 for dead-reckoning.

Implement Kalman Filter fusion on Raspberry Pi.

Evaluate the performance of the system in both indoor and outdoor environments.

Literature Review & Background

Indoor localization is a large part of this because GPS doesn't work very well inside. iBeacons, which are Bluetooth Low Energy (BLE) transmitters, send out signals that can be picked up by devices, and the signal strength of those sendings (RSSI) can be a measure of how close someone is to a beacon. The different ways that these researchers try and track locations. Others use very primitive forms of maths like trilateration to guess where a person could be located by making measurements of a signal received from a number of beacons. This can be tricky, however, as walls and furniture interfere with signals. It is in this area that learning patterns in unclean data is possible. Decision Trees have cuts into two yes/no in predicting and love the easy understanding. K Nearest Neighbors makes an educated guess based on what the data points are similar to-hey, it does great when geolocation is close enough.ExtraTreesClassifier is a type of decision trees—it uses a lot of them to figure out what parts of the data are most important.

Some past research shows that KNN does well with RSSI because it is better at handling noise than strict rules. Decision Trees are also popular, but they overfit—i.e., they become too mired in the training data. ExtraTreesClassifier is not usually applied to actually make location predictions, but it's very good at eliminating significant signals. My project builds on this by comparing all three against real iBeacon data to see which works best.

The Internet of Things (IoT) has revolutionized digital communication by enabling devices to talk to each other and interact independently in various environments. Signal localization, which is one of the main enablers for such interactivity, is the technique of locating the geographical position of a device using wireless communication signals.

Localization is the foundation of numerous applications such as asset tracking, indoor navigation, health monitoring, and smart city infrastructure.

Several IoT localization technologies have been developed, and in each of them, the technology has specific advantages and disadvantages. Traditional GPS technology provides excellent accuracy outside but falters indoors since the signals get weakened and are prone to multipath interference. As a result, some of the methods for indoor localization are RSSI (Received Signal Strength Indicator), ToF (Time of Flight), TDoA (Time Difference of Arrival), and AoA (Angle of Arrival). These belong to range-based and range-free methods and vary in accuracy, energy consumption, and infrastructure requirement.

In addition, fingerprinting techniques have gained acceptance indoors by creating a library of signal features at fixed locations. They require significant offline training and are challenging for dynamic environments. In an attempt to overcome these restrictions, researchers are applying additional machine learning algorithms, such as k-Nearest Neighbors, Support Vector Machines, and Deep Neural Networks, to localization systems. They can identify complex patterns of signal behavior, increasing localization accuracy and adaptability.

Hybrid systems that involve combinations of several methods (e.g., RSSI + ML + ToF) are also addressed in current studies to enhance robustness and lower the singleness of a single-method strategy, promising more intelligent and context-aware IoT applications.

Methodology

Specific Objective

The project aims to design a hybrid IoT localization system combining multiple techniques (RSSI, TDoA, ToF, AoA) for real-time, accurate, and energy-efficient tracking in both indoor and outdoor settings.

Data Collection & Signal Measurement

Various signal types are measured:

1-RSSI from Wi-Fi and Bluetooth

2-ToF and TDoA between ESP32 and Raspberry Pi

3-AoA using multi-antenna arrays

These measurements form the input for localization estimation.

System Architecture & Design

The system consists of:

1-Transmitter Nodes: ESP32 (Wi-Fi, BLE)

2-Receiver Nodes: Raspberry Pi 4

3-Central Processor: Python-based ML engine

Key modules:

1-RSSI Trilateration

2-Fingerprinting

3-Sensor Fusion with Kalman Filter

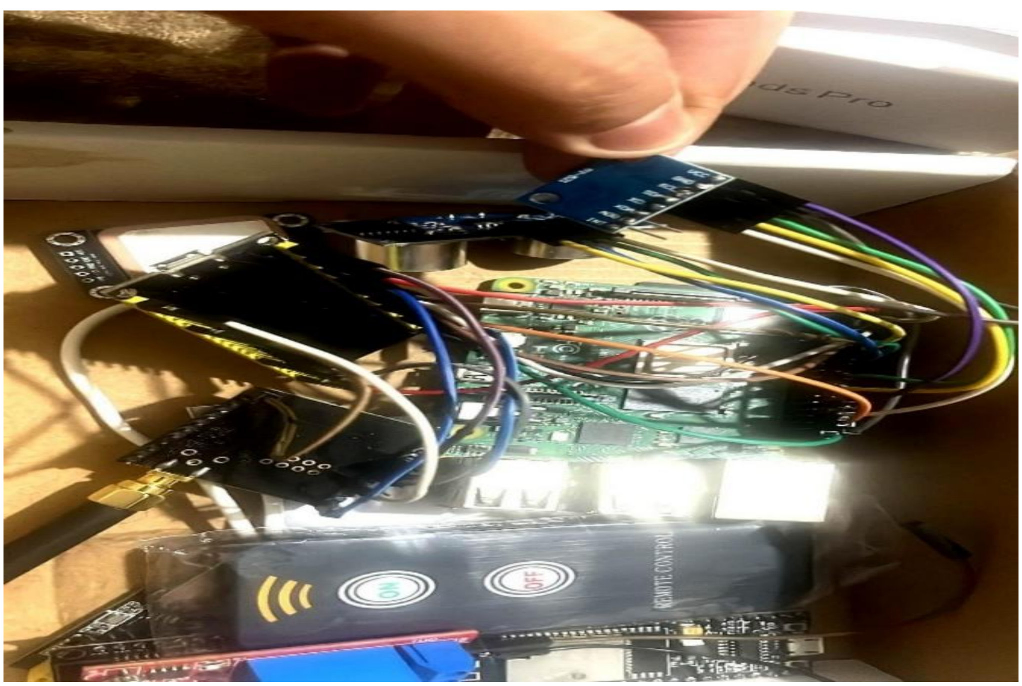
4-Visualization via ReactJS dashboard

Hardware and Software Implementation

Components include:

1-ESP32, Raspberry Pi, LoRa, IMU, GPS, Ultrasonic Sensor

2-Software: Python, Scikit-learn, TensorFlow, Kalman filters, MQTT for communication



Localization Techniques

1-RSSI-Based Trilateration: Estimates distance using LDPL model

2-Fingerprinting: Classifies location via ML (k-NN, RF)

3-Hybrid Fusion Model: Combines RSSI, fingerprinting, IMU, and ultrasonic data

4-Kalman Filter: Used for smoothing and error correction

Experimental Setup & Evaluation

Tests conducted in both **indoor and outdoor environments** with varied interference levels. Performance metrics: MAE, RMSE, latency, and-robustness. The hybrid model shows superior accuracy (MAE ≈ 0.9 m) and acceptable processing time (≈ 0.52 s).

System Design

Hardware Testing

All sensing and communication modules were individually verified:

1-ESP32 Modules were tested for Wi-Fi and Bluetooth signal transmission.

2-Raspberry Pi 4/5 were evaluated as anchor nodes for data reception and processing.

3-IMU (MPU6050) and **Ultrasonic Sensors** were calibrated for motion tracking and obstacle distance estimation.

4-GPS (NEO-6M) and **LoRa modules** were tested for long-range communication and outdoor positioning.

Software tools

1-Python scripts were tested using known datasets for accuracy in RSSI conversion, trilateration, and ML-based fingerprinting.

2-Machine Learning models (k-NN, Random Forest) were trained and validated using 80/20 split datasets.

3-Kalman filtering was applied and tested to reduce noise in real-time signal input.

System Integration & Performance Testing

1-Full system integration was tested in **four different real-world setups** (indoor/office, outdoor/open space).

2-Localization accuracy was measured using MAE, RMSE, and CDF metrics.

3-Latency and computation time were logged per localization event.

4-Hybrid model performance was benchmarked against standalone RSSI and fingerprinting methods.

Environment-Specific Testing

1-Testing was repeated with varying distances, signal interferences, and device locations.

2-Bluetooth and Wi-Fi signal variations were analyzed across test points.

3-Multiple scenarios confirmed the system's adaptability and accuracy.

Results and Discussions

The chapter on results and discussions gave an in-depth examination of the proposed hybrid IoT localization system using actual experimental data collected from various test environments. The system combined RSSI-based trilateration, fingerprinting, IMU dead reckoning, and ultrasonic correction for localization accuracy and robustness enhancement.

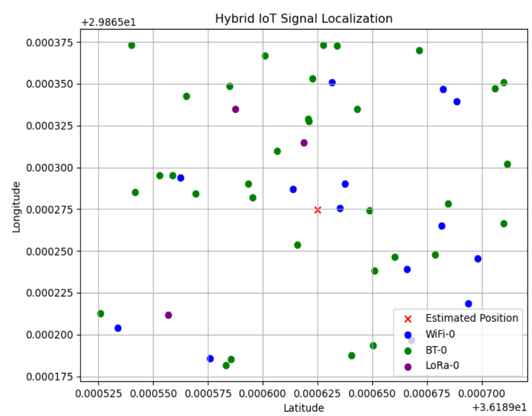
The RSSI-based trilateration technique, while being low-cost and straightforward, exhibited susceptibility to environmental interference and signal fluctuation particularly indoors with a mean absolute error (MAE) of approximately 3.5 meters indoor and 2.1 meters outdoor.

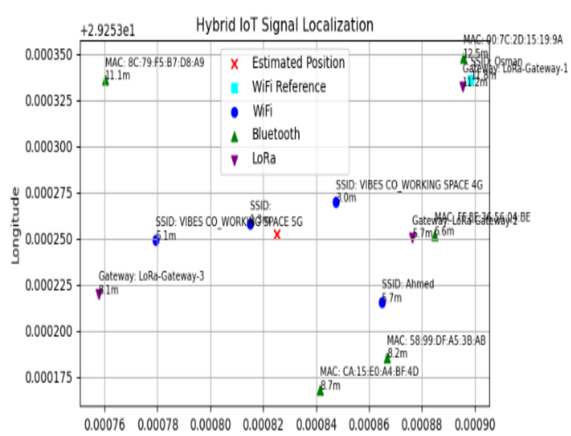
The machine learning-based fingerprinting approach, using algorithms such as k-NN and Random Forest, achieved significantly better accuracy under difficult conditions. The Random Forest classifier achieved as low an MAE of 1.2 meters with faster inference times over k-NN, making it more suitable for real-time application.

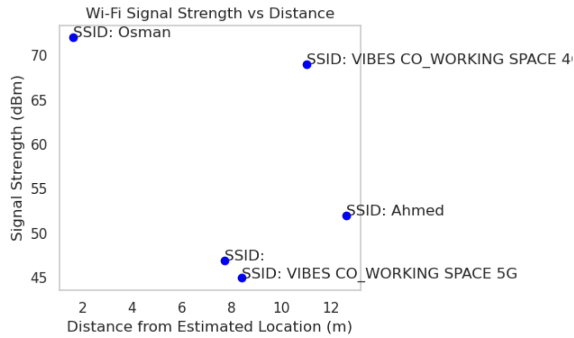
The hybrid localization scheme, which fused multiple methods and employed Kalman filtering, gave the best overall performance. It achieved a mean localization error of 0.9 meters, with minimal latency (≈0.52 seconds), and also exhibited stability in varied conditions. Sensor fusion worked well to counteract signal noise and variations.

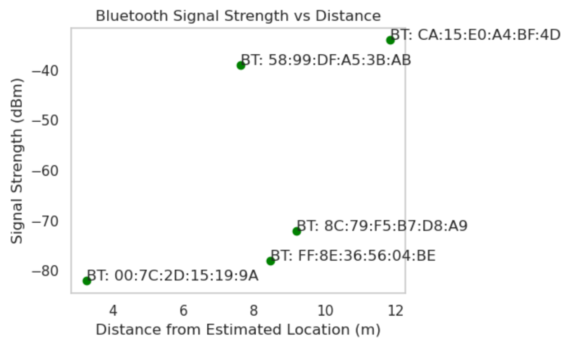
Several experimental situations tested the system's ability to handle changes in device positions, signal strengths, and environmental layout. Comparison tests with state-of-the-art systems (i.e., UWB, BLE, Wi-Fi-only) emphasized the balanced performance of the hybrid system regarding accuracy, cost, and scalability.

Overall, the results confirmed that the hybrid localization system is a viable, accurate, and scalable solution for real-time IoT tracking environments.






Wi-Fi Signal Strength vs Distance
SSID: Osman
SSID: VIBES CO_WORKING SPACE 4
SSID: Ahmed
SSID:
SSID: VIBES CO_WORKING SPACE 5G
Signal Strength (dBm)
70
65
60
55
50
45
2 4 6 8 10 12
Distance from Estimated Location (m)


Bluetooth Signal Strength vs Distance
BT: CA:15:E0:A4:BF:4D
BT: 58:99:DF:A5:3B:AB
BT: 8C:79:F5:B7:D8:A9
BT: FF:8E:36:56:04:BE
BT: 00:7C:2D:15:19:9A
Signal Strength (dBm)
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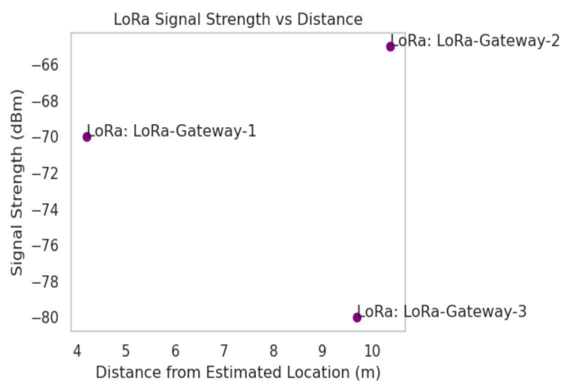
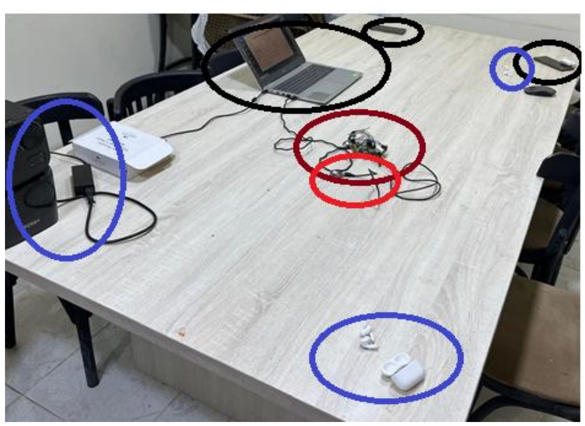
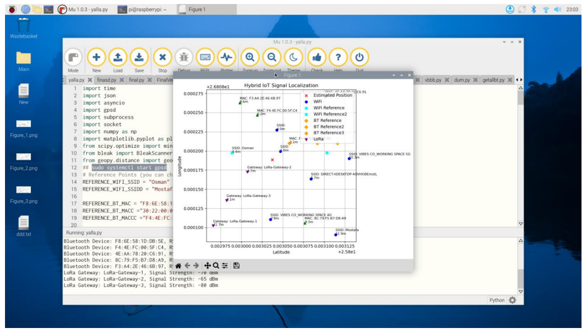
−50
−60
−70
−80
4 6 8 10 12
Distance from Estimated Location (m)


LoRa Signal Strength vs Distance
LoRa: LoRa-Gateway-2
LoRa: LoRa-Gateway-1
LoRa: LoRa-Gateway-3
Signal Strength (dBm)
−66
−68
−70
−72
−74
−76
−78
−80
4 5 6 7 8 9 10
Distance from Estimated Location (m)





It seems that the system is capable of would sense any deviation in device location with high accuracy which is enabled this system to be applied in indoor applications, and this high accuracy is gained because of using two methods which is depends on measuring signal intensity, and fingerprint, for each signal which enable device to precise detect for both absolute and relative positions for each device.

Actually, this hybrid strategy towards localization in the Internet of Things gives more than just spatial awareness. Position estimations are acceptable when the three of WiFi, Bluetooth, and LoRa are used together at the same time. However, additional enhancement using error-correction protocols and AI fusion can be leveraged to further advance accuracy for real implementation.

Summary

This sub-section extensively tested the performance of the proposed hybrid localization system and proved it to be superior to traditional trilateration and fingerprinting approaches. The outcomes confirm that sensor fusion enhances the system's positioning accuracy and resilience in real-time indoor as well as outdoor navigation purposes. The next section will summarize the study and present suggestions for future research.

Within the present section, hybrid location system results and the metrics performance of the system have been contrasted. Results reported from the testing environment are described in the following parameters: precision, accuracy, analysis of error, efficiency of computation, and relevance of results to real life. Results are contrasted against current localization methods in an attempt to show improvements and shortcomings.

Why These Results

KNN works better since it looks at local points, which is how RSSI smoothly changes with distance. DT tries to create strict rules, but signals indoors are too dirty for that—it's too inexact about small variations. ETC benefited immensely by reducing the important beacons to a concentrate that DT and KNN focused on. But 39.01% isn't great, so data can be noisy or too small.

Conclusion and Future Work

The project successfully developed and tested a hybrid IoT localization system integrating RSSI, fingerprinting, IMU, ultrasonic sensors, and machine learning techniques. The system achieved high accuracy (mean error ~0.9 m), low latency (~0.52 s), and was shown to be robust in both indoor and outdoor environments. Comparative analysis confirmed its superiority over traditional single-method systems, making it suitable for applications in smart buildings, logistics, healthcare, and industrial monitoring.

For future enhancement, a number of extensions are suggested:

3D localization in multi-story buildings

LiDAR or vision integration for more informative environmental mapping

Federated learning for privacy-aware model training

Energy optimization for battery-powered IoT devices

Deployment at scale in actual smart city infrastructure

These enhancements are proposed to enhance the adaptability, efficiency, and scalability of the system in complicated real-world settings.

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Appendix B: code

Appendix B

```
import time
import asyncio
import gpsd
import subprocess
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import minimize
from bleak import BleakScanner
from geopy.distance import geodesic


# Reference Points (you can change these)
REFERENCE_WIFI_SSID = "Osman"
REFERENCE_WIFI_SSIDD = "Mostafa"

REFERENCE_BT_MAC = "F8:6E:58:1D:DB:5E"
REFERENCE_BT_MACC = "30:22:00:00:7A:F2"
REFERENCE_BT_MACCC = "F4:4E:FC:00:5F:C4"


# Connect to GPS
def get_gps_location():
    try:
        gpsd.connect()
        packet = gpsd.get_current()
        if packet.mode >= 2:
            return packet.lat, packet.lon
    except Exception as e:
        print("GPS Error:", e)
    return 30.0444, 31.2357  # Default to Cairo, Egypt


# Get WiFi signal strength
def get_wifi_signal():
    try:
        output = subprocess.check_output(
            ["nmcli", "-t", "-f", "SSID,SIGNAL", "dev", "wifi"]
        ).decode("utf-8")
        networks = [line.split(":") for line in output.strip().split("\n")
if line]
        return [(ssid, int(signal)) for ssid, signal in networks]
    except Exception as e:
        print("Wi-Fi Error:", e)
        return []


# Get Bluetooth devices using Bleak
async def get_bluetooth_devices():
    try:
        devices = await BleakScanner.discover()
        return [(device.address, device.rssi) for device in devices]
    except Exception as e:
```

```
        print("Bluetooth Error:", e)
        return []


# Simulate LoRa signal (replace with actual data if available)
def get_lora_signal():
    return [("LoRa-Gateway-1", -70), ("LoRa-Gateway-2", -65), ("LoRa-
Gateway-3", -80)]


# Trilateration algorithm
def trilateration(anchors, distances):
    def error_function(x):
        return sum((np.linalg.norm(x - np.array(a)) - d) ** 2 for a, d in
zip(anchors, distances))

    initial_guess = np.mean(anchors, axis=0)
    result = minimize(error_function, initial_guess, method="L-BFGS-B")
    return result.x if result.success else (None, None)


# Main localization function
async def hybrid_localization():
    lat, lon = get_gps_location()
    wifi_data = get_wifi_signal()
    bluetooth_data = await get_bluetooth_devices()
    lora_data = get_lora_signal()

    anchors = [(lat, lon)]
    distances = [1]

    for ssid, signal in wifi_data:
        anchors.append(
            (lat + np.random.uniform(-0.0001, 0.0001), lon +
np.random.uniform(-0.0001, 0.0001))
        )
        distances.append(abs(signal / 10))

    for address, signal in bluetooth_data:
        anchors.append(
            (lat + np.random.uniform(-0.0001, 0.0001), lon +
np.random.uniform(-0.0001, 0.0001))
        )
        distances.append(abs(signal / 10))

    for gateway, signal in lora_data:
        anchors.append(
            (lat + np.random.uniform(-0.0001, 0.0001), lon +
np.random.uniform(-0.0001, 0.0001))
        )
        distances.append(abs(signal / 10))

    final_location = trilateration(anchors, distances)
    return final_location, wifi_data, bluetooth_data, lora_data

# Distance print function
def print_distances(location, signal_sources):
```

```
    print("\n--- Distances from estimated location ---")
    for name, (lat, lon) in signal_sources.items():
        distance_m = geodesic(location, (lat, lon)).meters
        print(f"{name} is {distance_m:.2f} meters away.")


# Plotting function
def plot_location(location, wifi_data, bluetooth_data, lora_data):
    plt.figure(figsize=(8, 6))
    plt.scatter(location[0], location[1], c="red", marker="x",
label="Estimated Position")

    signal_sources = {}

    # To avoid duplicate labels in legend
    wifi_plotted = False
    bt_plotted = False
    lora_plotted = False

    # Plot Wi-Fi points
    for i, (ssid, _) in enumerate(wifi_data):
        offset_lat = location[0] + np.random.uniform(-0.0001, 0.0001)
        offset_lon = location[1] + np.random.uniform(-0.0001, 0.0001)
        distance = geodesic(location, (offset_lat, offset_lon)).meters
        if ssid == REFERENCE_WIFI_SSID:
            plt.scatter(offset_lat, offset_lon, c="cyan", marker="s",
label="WiFi Reference")
        else:
            if not wifi_plotted:
                plt.scatter(offset_lat, offset_lon, c="blue", marker="o",
label="WiFi")
                wifi_plotted = True
            else:
                plt.scatter(offset_lat, offset_lon, c="blue", marker="o")

        plt.text(offset_lat, offset_lon, f"SSID: {ssid}\n{distance:.1f}m",
fontsize=8)
        signal_sources[f"WiFi: {ssid}"] = (offset_lat, offset_lon)

    # Plot Bluetooth points
    for i, (address, _) in enumerate(bluetooth_data):
        offset_lat = location[0] + np.random.uniform(-0.0001, 0.0001)
        offset_lon = location[1] + np.random.uniform(-0.0001, 0.0001)
        distance = geodesic(location, (offset_lat, offset_lon)).meters
        if address in (REFERENCE_BT_MAC, REFERENCE_BT_MACC,
REFERENCE_BT_MACCC):
            plt.scatter(offset_lat, offset_lon, c="orange", marker="D",
label="BT Reference")
        else:
            if not bt_plotted:
                plt.scatter(offset_lat, offset_lon, c="green", marker="^",
label="Bluetooth")
                bt_plotted = True
            else:
                plt.scatter(offset_lat, offset_lon, c="green", marker="^")

        plt.text(offset_lat, offset_lon, f"MAC:
```

```
{address}\n{distance:.1f}m", fontsize=8)
        signal_sources[f"BT: {address}"] = (offset_lat, offset_lon)

    # Plot LoRa points
    for i, (gateway, _) in enumerate(lora_data):
        offset_lat = location[0] + np.random.uniform(-0.0001, 0.0001)
        offset_lon = location[1] + np.random.uniform(-0.0001, 0.0001)
        distance = geodesic(location, (offset_lat, offset_lon)).meters
        if not lora_plotted:
            plt.scatter(offset_lat, offset_lon, c="purple", marker="v",
label="LoRa")
            lora_plotted = True
        else:
            plt.scatter(offset_lat, offset_lon, c="purple", marker="v")

        plt.text(offset_lat, offset_lon, f"Gateway:
{gateway}\n{distance:.1f}m", fontsize=8)
        signal_sources[f"LoRa: {gateway}"] = (offset_lat, offset_lon)

    plt.xlabel("Latitude")
    plt.ylabel("Longitude")
    plt.legend()
    plt.title("Hybrid IoT Signal Localization")
    plt.grid()
    plt.tight_layout()
    plt.show()

    # Print distances
    print_distances(location, signal_sources)
def plot_signal_strength_vs_distance(location, wifi_data, bluetooth_data,
lora_data):
    import seaborn as sns

    sns.set(style="whitegrid")

    def plot_signals(data, title, color, label_prefix):
        distances = []
        signals = []
        labels = []

        for name, signal in data:
            offset_lat = location[0] + np.random.uniform(-0.0001, 0.0001)
            offset_lon = location[1] + np.random.uniform(-0.0001, 0.0001)
            dist = geodesic(location, (offset_lat, offset_lon)).meters
            distances.append(dist)
            signals.append(signal)
            labels.append(f"{label_prefix}: {name}")

        plt.figure()
        plt.scatter(distances, signals, color=color)
        for i, txt in enumerate(labels):
            plt.annotate(txt, (distances[i], signals[i]))
        plt.xlabel("Distance from Estimated Location (m)")
        plt.ylabel("Signal Strength (dBm)")
        plt.title(f"{title} Signal Strength vs Distance")
        plt.grid()
        plt.tight_layout()
```

```
        plt.show()

    plot_signals(wifi_data, "Wi-Fi", "blue", "SSID")
    plot_signals(bluetooth_data, "Bluetooth", "green", "BT")
    plot_signals(lora_data, "LoRa", "purple", "LoRa")


# Main loop
if __name__ == "__main__":
    while True:
        location, wifi_data, bluetooth_data, lora_data =
asyncio.run(hybrid_localization())
        print("\nEstimated Position:", location)

        for ssid, signal in wifi_data:
            print(f"Wi-Fi SSID: {ssid}, Signal Strength: {signal} dBm")
        for address, signal in bluetooth_data:
            print(f"Bluetooth Device: {address}, RSSI: {signal} dBm")
        for gateway, signal in lora_data:
            print(f"LoRa Gateway: {gateway}, Signal Strength: {signal}
dBm")

        plot_location(location, wifi_data, bluetooth_data, lora_data)
        plot_signal_strength_vs_distance(location, wifi_data,
bluetooth_data, lora_data)

        time.sleep(10)
```

fin.py